

FMCG ERP

Full Reference Manual

Version: 1.0

Date: 5 June 2026

Scope: All Modules — Complete Reference

Chapters: 15

Screens may change as the ERP evolves. Always follow company SOPs and manager instructions.

Confidential — Internal Use Only

Full ERP Reference Manual — Index

Date: 2026-05-18

Version: 1.0

System: FMCG ERP — Kenya

This is the complete reference manual covering every workspace and module in the ERP. Unlike the Kenya go-live role manuals, this manual covers all pages regardless of role.

How to Use This Manual

- **Staff training:** use the Kenya go-live role manuals in `../kenya-go-live/` instead
 - **Admin reference:** use this manual when setting up or investigating any module
 - **Developer reference:** see API docs at `/docs` and backend source
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Chapter Index

Chapter	Topic	Workspaces Covered
01	Dashboard & Navigation	/dashboard, sidebar, search
02	Master Data	products, materials, suppliers, recipes, BOM
03	Procurement	procurement, suppliers
04	Inventory & Warehouse	inventory, warehouses, wms
05	Production	production, shop-floor, planning
06	Quality & Compliance	quality, compliance
07	Sales & Distribution	sales, logistics, crm, marketing, pos
08	Finance	finance, payroll
09	HR & Payroll	hr, payroll
10	Admin & Security	admin, integrations, logs, approvals
11	AI & Automation	ai, analytics
12	Reports & Exports	analytics, all workspace export functions
13	Standalone Pages	npd, maintenance, utility-management, helpdesk, documents, communication
14	Old Route Compatibility	All /dashboard/* redirects

Workspace Quick Reference

Workspace	URL	Role	Key Tabs
Dashboard	/dashboard	All	—
Products	/dashboard/products	Admin/Prod	products
Materials	/dashboard/materials	Warehouse	materials
Suppliers	/dashboard/suppliers	Procurement	suppliers
Inventory	/dashboard/inventory	Warehouse	stock, movements, shelf-life, cycle-count, traceability, serials, valuation
Warehouses	/dashboard/warehouses	Warehouse	warehouses, wms
WMS	/dashboard/wms	Warehouse	zones, locations, quarantine
Procurement	/dashboard/procurement	Procurement	purchase-requests, orders, rfq, deliveries, suppliers, suggestions, subcontracting, landed-cost
Production	/dashboard/production	Production	plans, orders, scheduling, execution, material-flow, batch-lots, quality-control, oee, downtime, waste-yield, costing
Shop Floor	/dashboard/shop-floor	Production	terminal, supervisor, queue, downtime, handover
BOM	/dashboard/bom	Production	list, substitutes, compare, conversion
Recipes	/dashboard/recipes	Production	recipes
Planning	/dashboard/planning	Production	mrp, mps, capacity, advanced, kanban
NPD	/dashboard/npd	Production	npd
Quality	/dashboard/quality	Quality	inspections, certificates, parameters, consumer-complaints, qms, allergen, brand-assets
Compliance	/dashboard/compliance	Quality	gs1, regulatory-certs
Sales	/dashboard/sales	Sales	orders, invoices, customers, quotes, shipments, van-sales, collections, returns, price-lists, field-sales
Logistics	/dashboard/logistics	Sales	shipments, containers, arrivals, fleet

Workspace	URL	Role	Key Tabs
CRM	/dashboard/crm	Sales	pipeline, leads, opportunities, loyalty, nps, surveys
Marketing	/dashboard/marketing	Sales	campaigns, promotions, tpm, market-intel
POS	/dashboard/pos	Sales	pos, sales, sessions
Finance	/dashboard/finance	Finance	accounting, bank-recon, fixed-assets, receivables, cashbook, budget, mpesa, tax, expenses
HR	/dashboard/hr	HR	employees, attendance, payroll, leave, recruitment, appraisals, training, expenses
Payroll	/dashboard/payroll	HR	overview, profiles, reports
Admin	/dashboard/admin	Admin	users, roles, permissions, companies, security, approvals, system-config, logs
Integrations	/dashboard/integrations	Admin	mpesa, sync, webhooks, developer
Analytics	/dashboard/analytics	Manager	overview, sales, production, inventory, finance, report-builder
AI	/dashboard/ai	All	chat, predictions, formulations
Documents	/dashboard/documents	All	documents, compliance, knowledge-base, esign
Communication	/dashboard/communication	All	messages, chatter, calendar, notifications
Helpdesk	/dashboard/helpdesk	All	all, open, escalated, sla
Maintenance	/dashboard/maintenance	Production	assets, breakdowns, plans, predictive
Utilities	/dashboard/utility-management	Production	electricity, water, steam, solar, esg
Logs	/dashboard/logs	Admin	operational, mpesa, security
Approvals	/dashboard/approvals	Manager	all, rules

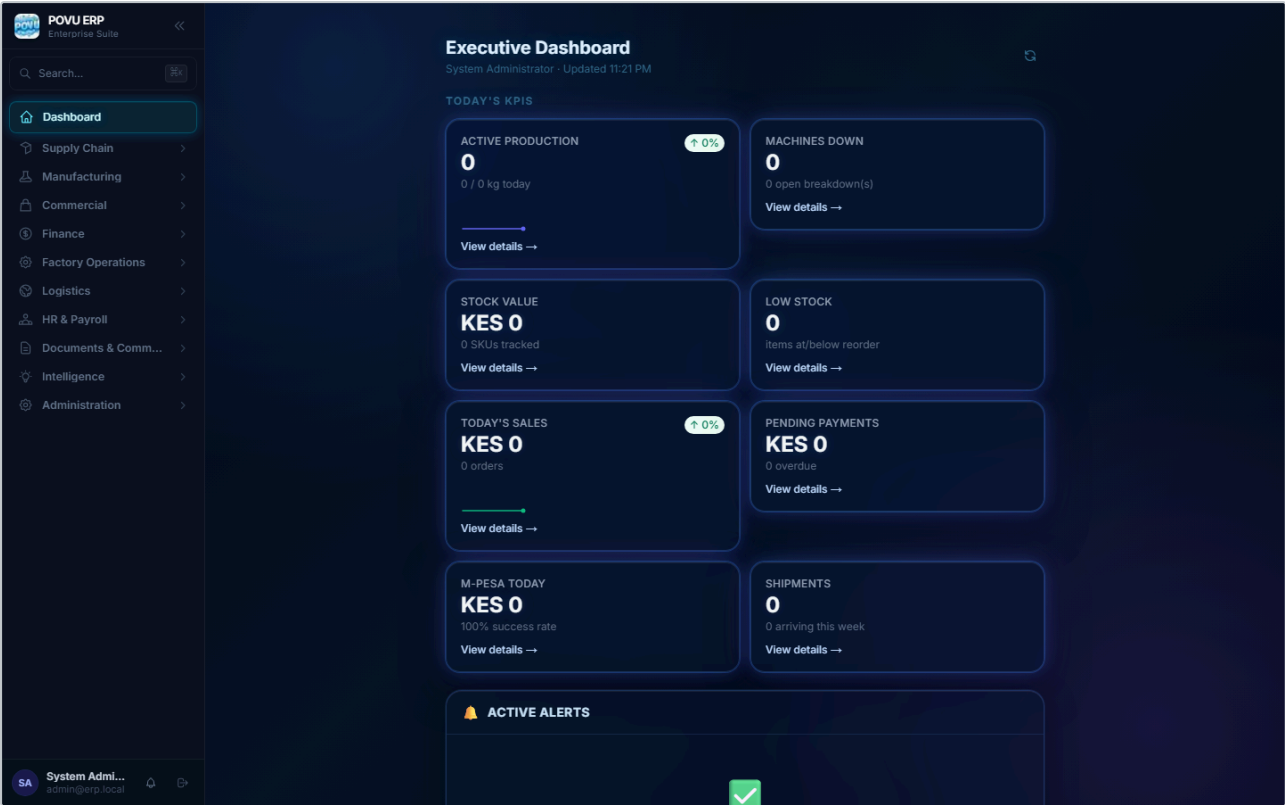
Dashboard and Navigation

URL: /dashboard

Module: Core

Permission: Any authenticated user

Screenshot



Purpose

The dashboard is the home page after login. It shows business KPIs, pending tasks, and quick navigation to all modules.

Sidebar Navigation

The left sidebar shows all accessible modules, grouped by cluster:

Cluster	Workspaces
Supply Chain	Products, Materials, Suppliers, Inventory, Warehouses, Procurement
Manufacturing	Production, Planning, NPD, BOM, Recipes, Quality, Compliance, Shop Floor
Commercial	Sales, CRM, Marketing, POS
Finance	Finance
Factory Operations	Maintenance, Utilities
Logistics	Logistics
HR & Payroll	HR, Kenya Payroll
Documents & Communication	Documents, Communication, Helpdesk
Intelligence	AI, Analytics
Administration	Admin, Integrations

Workspaces you do not have permission for are hidden or greyed out.

Global Search

Press `Ctrl+K` (or click the search icon) to open global search. Search for:

- Module names (e.g., "production")
- Page tab names (e.g., "cycle count")
- Records (if backend search is enabled)

Search hints from workspace `searchHints` are shown in the dropdown.

Theme

Toggle dark/light mode via the user menu (top right).

User Menu

Top-right corner:

- Profile: view/edit your profile
- Security: change password, set up 2FA

- Notifications: unread notification count
 - Logout
-

Navigation Pattern

All modules use the workspace pattern:

1. Click a workspace in the sidebar (e.g., Production)
2. Workspace opens with a default tab active
3. Click tab buttons at the top to switch between functions
4. Tab state persists in the URL: `/dashboard/production?tab=orders`

Bookmarking a tab URL opens the correct tab directly.

Master Data

URLs: `/dashboard/products`, `/dashboard/materials`, `/dashboard/suppliers`,
`/dashboard/recipes`, `/dashboard/bom`

Module: Master Data

Permission: `products.view`, `procurement.view`, `recipe.view`, `bom.view`

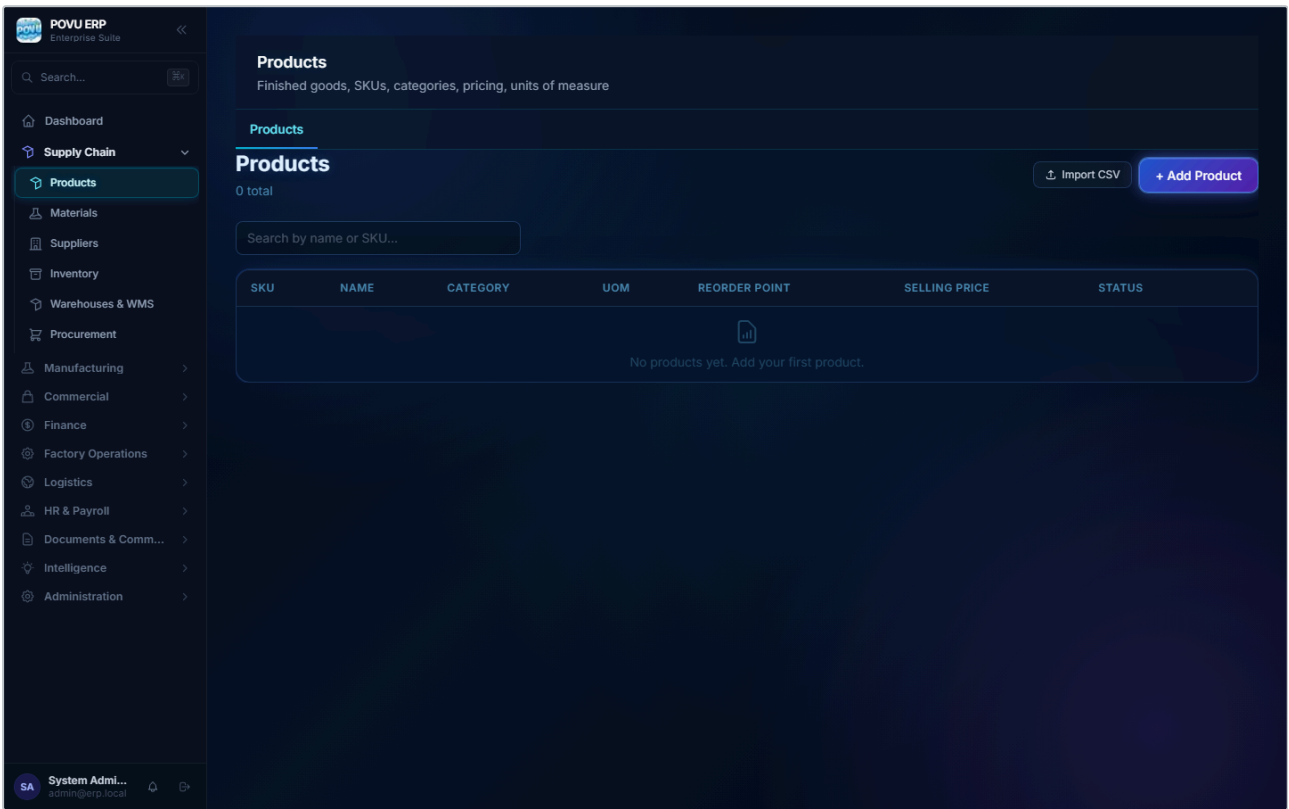
Overview

Master data is the foundation of the ERP. Products, materials, suppliers, recipes, and BOMs must be set up before any operational transactions can begin.

Products Workspace

URL: `/dashboard/products`

Tab: `products`



Finished goods catalogue. Each product has:

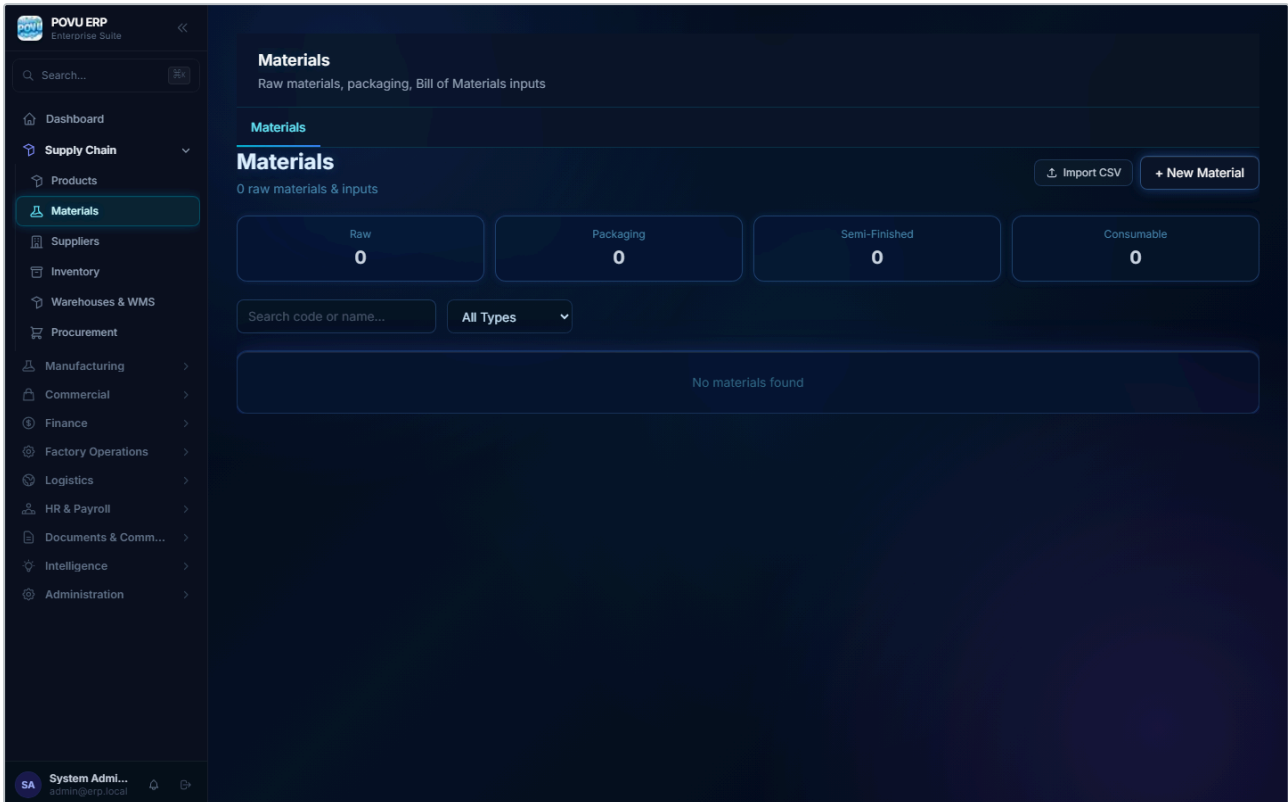
- SKU code, description, category
- Unit of measure (KG, Litre, Unit, Carton)
- Packaging specification
- Allergen flags

- Active BOM version link
- Costing group
- Price list assignment

Materials Workspace

URL: /dashboard/materials

Tab: materials



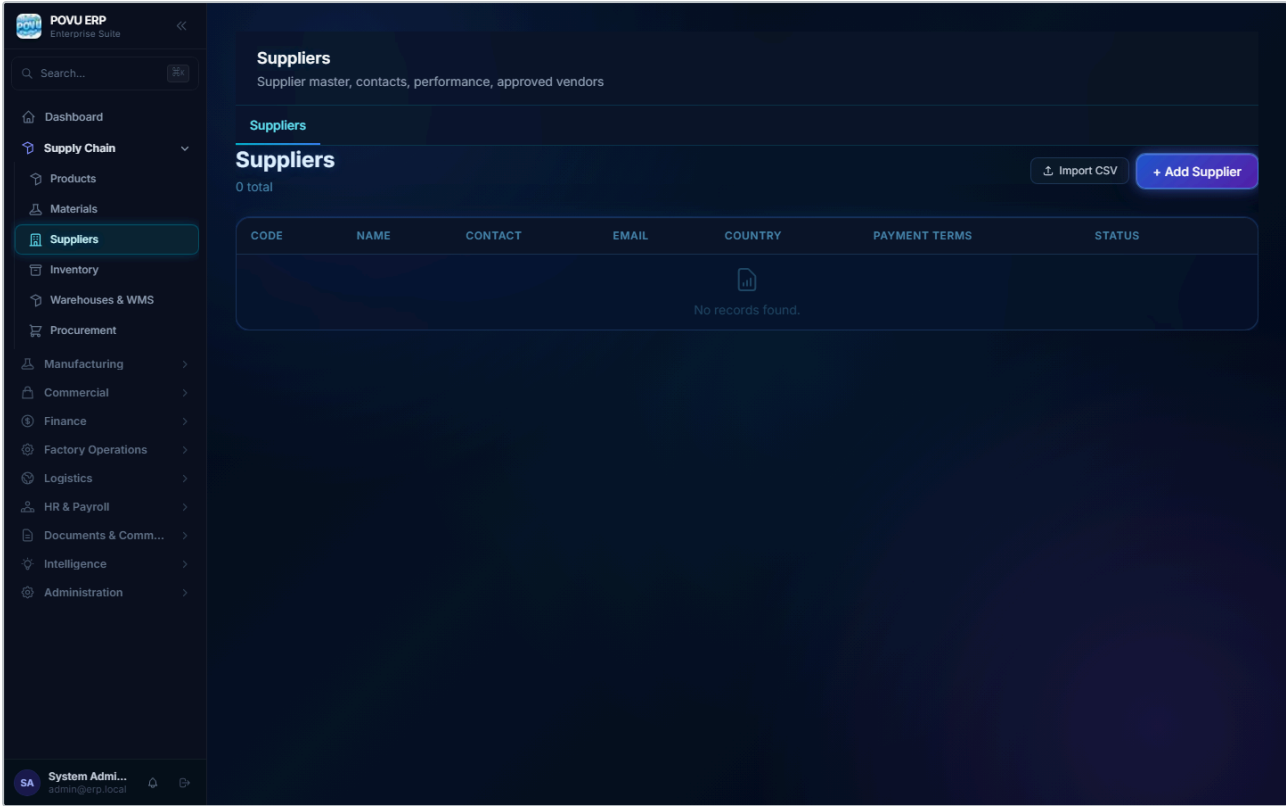
Raw material master. Each material has:

- Material code and description
- Category (raw material, packaging, indirect)
- Unit of measure
- Supplier links
- Reorder point and reorder quantity
- Lead time (days)
- Storage conditions
- Allergen content

Suppliers Workspace

URL: /dashboard/suppliers

Tab: suppliers

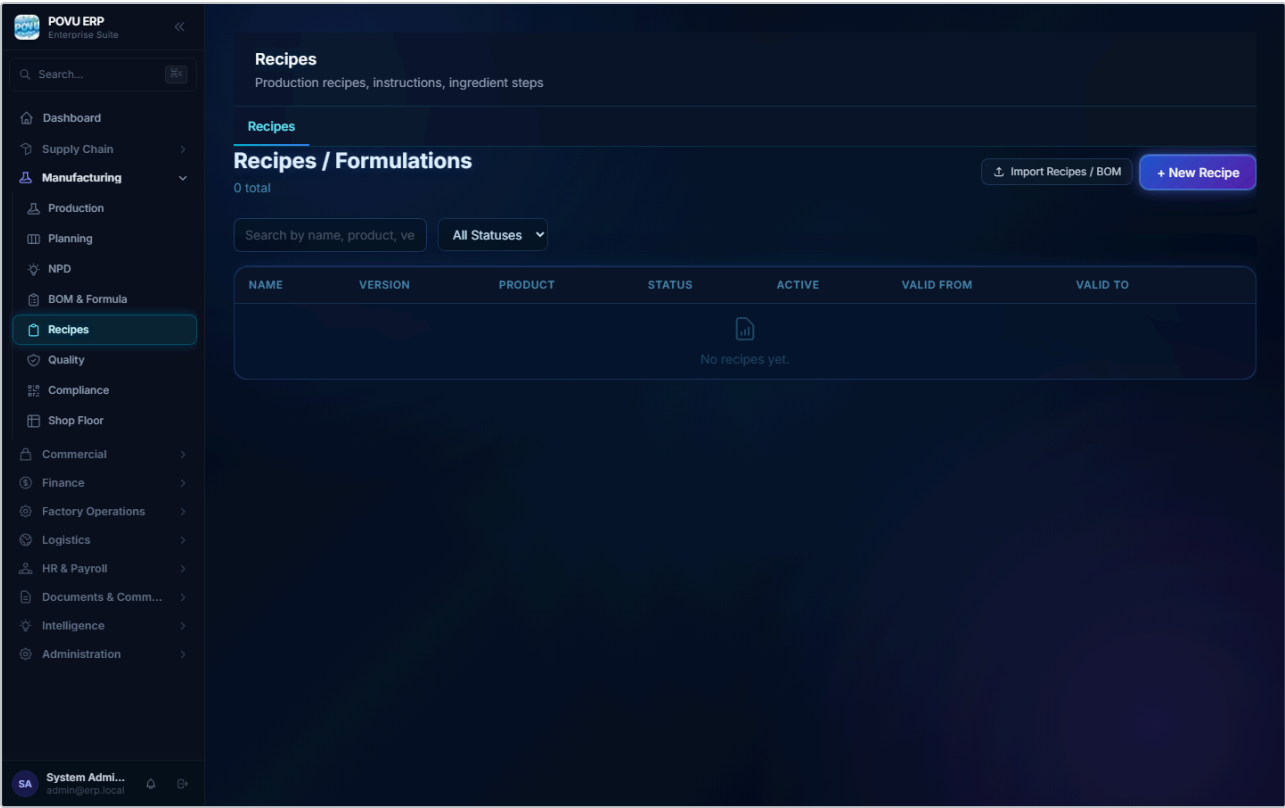


Supplier master used by Procurement. See Chapter 03 (Procurement) for full details.

Recipes Workspace

URL: /dashboard/recipes

Tab: recipes

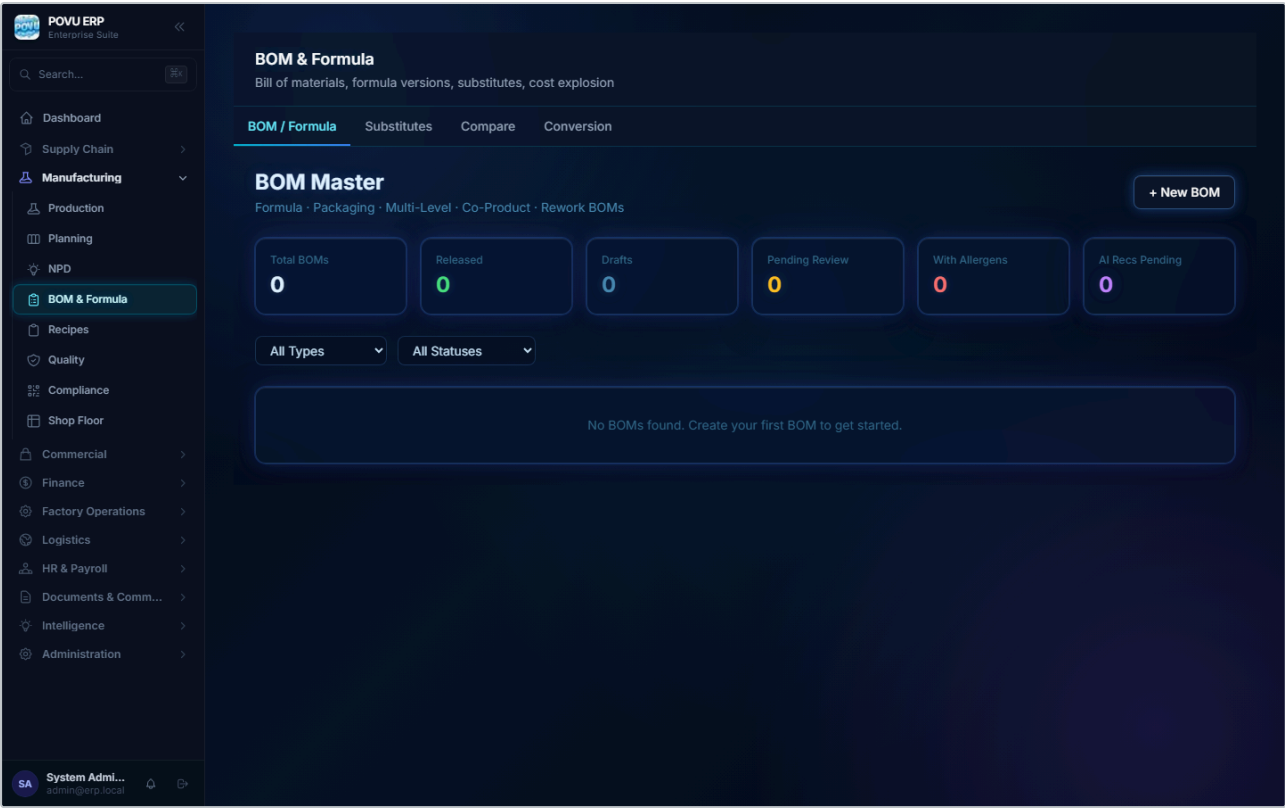


Product formulas. Recipes define ingredients, quantities per batch, and processing steps. A recipe becomes a BOM when linked to a product.

BOM & Formula Workspace

URL: /dashboard/bom

Tabs: list, substitutes, compare, conversion



BOM Tabs

Tab	Purpose
list	All BOM versions with effective dates
substitutes	Allowed material substitutions
compare	Side-by-side BOM version comparison
conversion	Batch size conversion profiles

BOM Version Control

Each BOM has:

- Version number
- Effective from / effective to date
- Status: Draft, Active, Archived

Only one BOM version should be Active for each product at any time.

Substitutes

When a material is unavailable:

1. BOM → Substitutes tab
2. Find the material

3. See allowed substitutes and quantity equivalence
 4. Production supervisor selects substitute when creating the production order
-

Setup Sequence

For a new product:

1. Create material records for all ingredients
2. Create the product in Products workspace
3. Create a recipe in Recipes workspace
4. Create a BOM in BOM workspace linking product to recipe
5. Set BOM status to Active with today as effective date
6. Test by creating a production order — verify material requirements

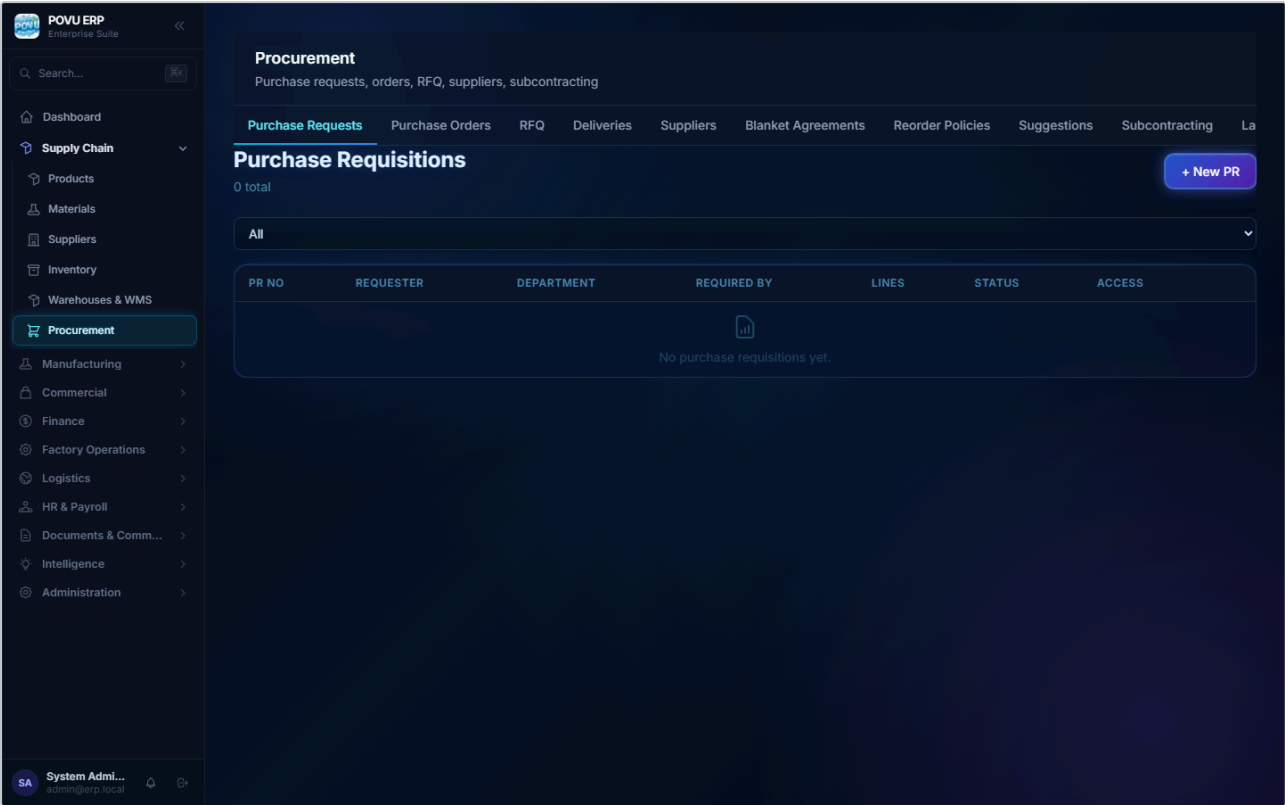
Procurement

URL: /dashboard/procurement

Module: Procurement

Permission: procurement.view

Screenshot



Tabs

Tab	URL	Purpose
Purchase Requests	?tab=purchase-requests	Internal purchase requisitions
Orders	?tab=orders	Purchase order management
RFQ	?tab=rfq	Request for Quotation
Deliveries	?tab=deliveries	Goods receipt notes
Suppliers	?tab=suppliers	Supplier performance
Blanket Agreements	?tab=blanket-agreements	Standing order contracts
Reorder Policies	?tab=reorder-policies	Auto-reorder rules
AI Suggestions	?tab=suggestions	AI-driven procurement suggestions
Subcontracting	?tab=subcontracting	Outsourced production orders
Landed Cost	?tab=landed-cost	Import cost allocation
Supplier Portal	?tab=supplier-portal	Portal for supplier self-service

PO Status Flow

Draft → Submitted → Approved → Sent to Supplier → Partially Received → Fully Received →

Approval Thresholds

Configure in Admin → Approvals:

- POs below threshold: auto-approved
- POs above threshold: require manager approval

Blanket Agreements

Standing arrangements with preferred suppliers for regular materials:

1. Set agreed price and quantity for a period
2. System creates POs against the agreement without re-quoting
3. Agreement expires when quantity or date limit reached

Landed Cost Allocation

For imported materials:

- Freight, insurance, customs duty, port charges
 - Allocate by: line value, weight, or quantity
 - System adds landed cost to inventory unit cost
-

Related Workspaces

- Suppliers (`/dashboard/suppliers`) — supplier master
- Inventory (`/dashboard/inventory`) — stock after receipt
- Finance (`/dashboard/finance`) — payables, invoice matching

Inventory and Warehouse

URLs: `/dashboard/inventory`, `/dashboard/warehouses`, `/dashboard/wms`

Module: Inventory / WMS

Permission: `inventory.view`

Screenshot

POVU ERP
Enterprise Suite

Search...

Dashboard

Supply Chain

Products

Materials

Suppliers

Inventory

Warehouses & WMS

Procurement

Manufacturing

Commercial

Finance

Factory Operations

Logistics

HR & Payroll

Documents & Comm...

Intelligence

Administration

Help Center

SA System Admin...
admin@erp.local

Inventory

Stock ledger, movements, cycle counts, shelf life, traceability

Stock Ledger

Movements

Cycle Count

Shelf Life

Traceability

Serials

Valuation

Inventory

5 stock positions

Current Stock

Stock Entry (IN)

Stock Issue (OUT)

Transfer

SKU

PRODUCT

WAREHOUSE

LOT

EXPIRY

ON HAND

AVAILABLE

RESERVED

ACTIONS

POVU-LD-1L

POVU Liquid Detergent 1L

FG-WH — Finished Goods Warehouse

no lot

—

4,500

4,500

0

Adjust Delete

POVU-FS-500ML

POVU Fabric Softener 500ml

FG-WH — Finished Goods Warehouse

no lot

—

2,800

2,800

0

Adjust Delete

POVU-DW-500ML

POVU Dishwashing Liquid 500ml

FG-WH — Finished Goods Warehouse

no lot

—

3,600

3,600

0

Adjust Delete

POVU-SC-750ML

POVU Surface Cleaner 750ml

FG-WH — Finished Goods Warehouse

no lot

—

2,300

2,300

0

Adjust Delete

POVU-HS-200ML

POVU Hand Sanitizer 200ml

FG-WH — Finished Goods Warehouse

no lot

—

7,500

7,500

0

Adjust Delete

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Inventory Workspace Tabs

Tab	URL	Purpose
Stock	?tab=stock	Current stock levels by SKU and location
Movements	?tab=movements	All in/out movements history
Cycle Count	?tab=cycle-count	Physical count scheduling
Shelf Life	?tab=shelf-life	FEFO expiry tracking
Traceability	?tab=traceability	Lot/batch forward/backward trace
Serials	?tab=serials	Serial number tracking
Valuation	?tab=valuation	Stock value by cost method

Warehouses Workspace Tabs

Tab	URL	Purpose
Warehouses	?tab=warehouses	Warehouse list and configuration
WMS	?tab=wms	WMS layout inside warehouses context

WMS Workspace Tabs

Tab	URL	Purpose
Zones	?tab=zones	Zone configuration
Locations	?tab=locations	Bin location list and capacity
Quarantine	?tab=quarantine	Quarantined stock

Key Concepts

Available stock = On Hand – Reserved (reserved by released production/sales orders)

FEFO (First Expired, First Out): The system suggests picking the batch with the nearest expiry date first. Critical for food production.

Lot/Batch traceability: Every stock movement carries the batch number. The traceability tab shows the full genealogy.

Valuation methods: FIFO, AVCO (Average Cost), or Standard Cost. Set per product category in Admin.

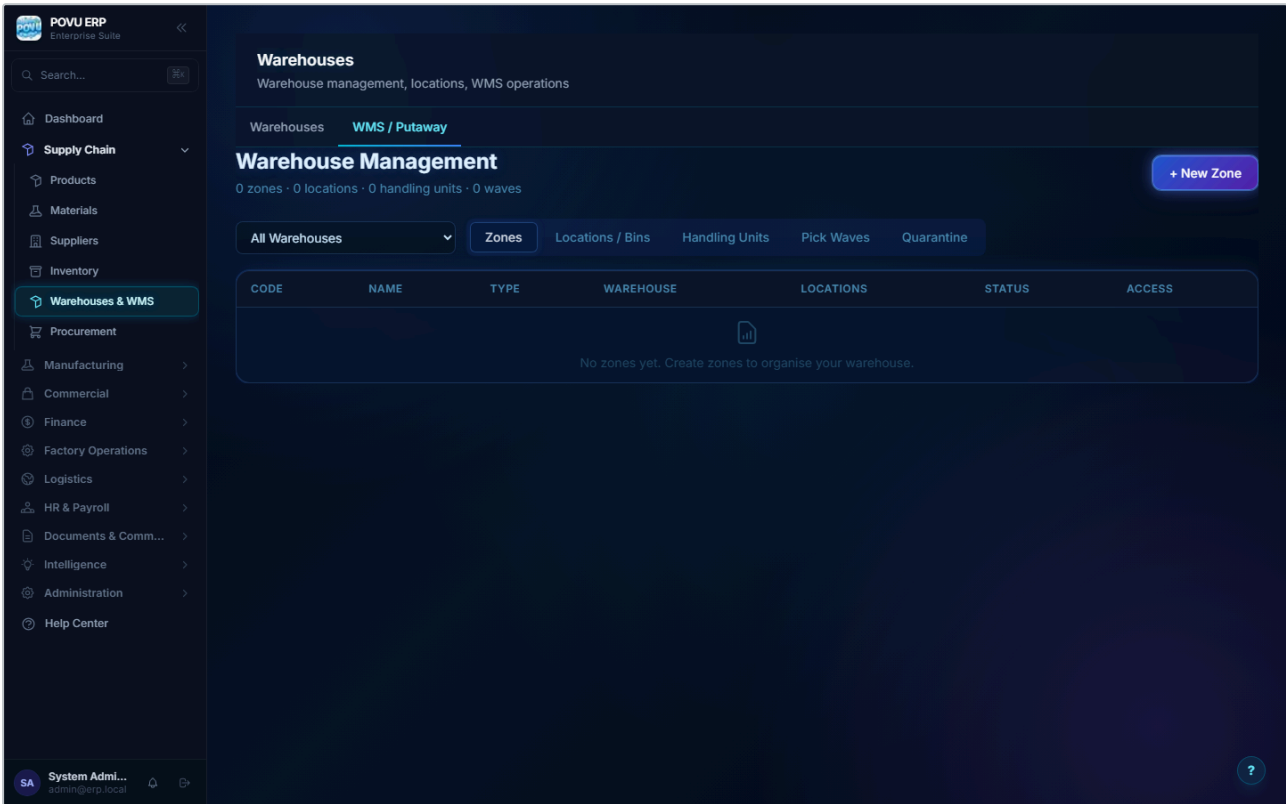
Quarantine Workflow

1. Incoming goods failing QC → marked Quarantine during receipt
2. WMS Quarantine tab shows all quarantined stock
3. QC technician inspects
4. Release to stock (if pass) or write off (if fail)
5. Write-off requires manager approval

Related Workspaces

- Procurement (`/dashboard/procurement`) — goods receipt
- Production (`/dashboard/production`) — material consumption
- Quality (`/dashboard/quality`) — QC inspections linked to batches

Screenshot



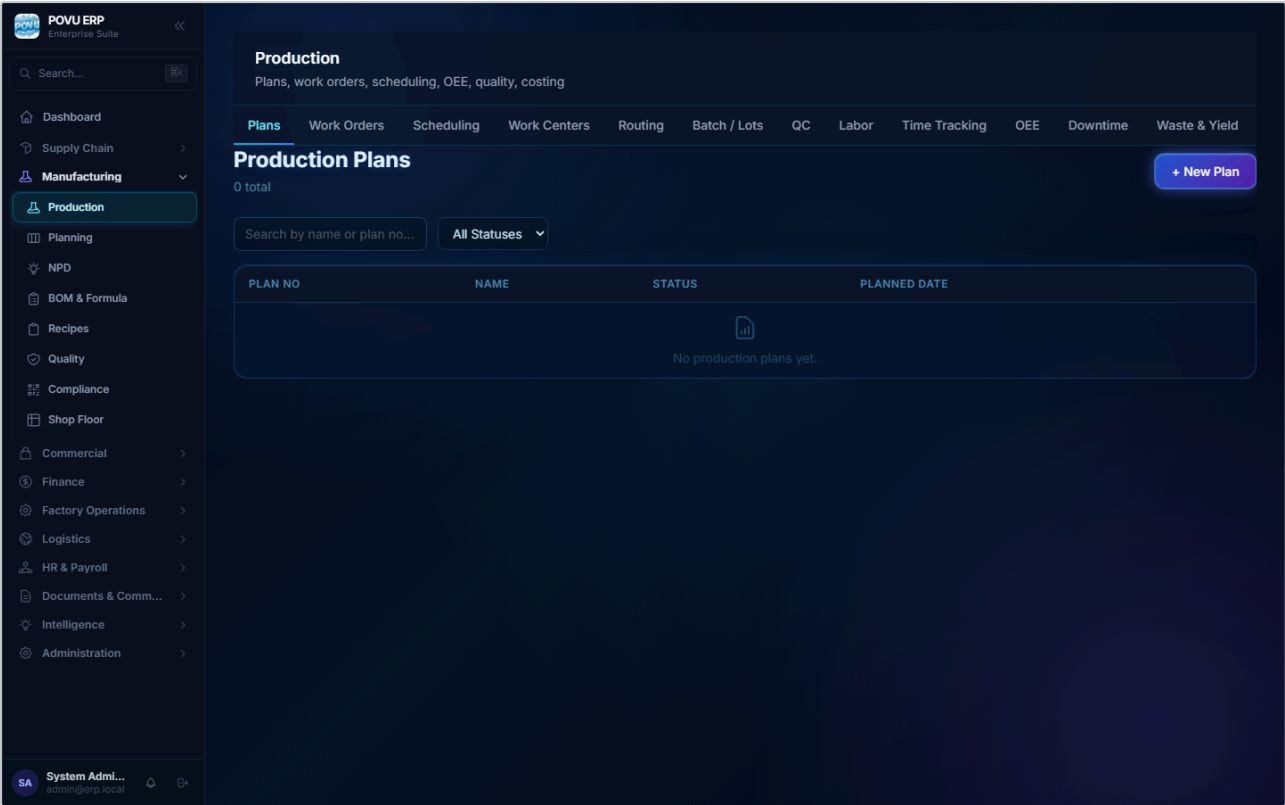
Production

URL: /dashboard/production

Module: Production

Permission: production.view

Screenshot



Purpose

The Production workspace manages the entire manufacturing lifecycle: planning, order execution, material consumption, quality control, batch tracking, OEE, and costing.

Tabs

Tab	URL	Purpose
Plans	?tab=plans	Daily/weekly production plans
Orders	?tab=orders	Production order management
Scheduling	?tab=scheduling	Finite scheduling board
Work Centers	?tab=work-centers	Machine/line configuration
Routing	?tab=routing	Operation sequences
Batch & Lots	?tab=batch-lots	Batch number assignment
Quality Control	?tab=quality-control	In-process QC
Labor	?tab=labor	Labor allocation
Time Tracking	?tab=time-tracking	Job timing
OEE	?tab=oeo	Overall Equipment Effectiveness
Downtime	?tab=downtime	Downtime log
Waste & Yield	?tab=waste-yield	Waste and yield recording
WIP	?tab=wip	Work in progress
Costing	?tab=costing	Standard vs actual cost
Variance	?tab=variance	Cost variance analysis
Reports	?tab=reports	Production reports
Execution	?tab=execution	Production output confirmation
Machine Ops	?tab=machine-ops	Machine operator view
Material Flow	?tab=material-flow	Material issue and backflush
Projects	?tab=projects	Production projects

Key Workflows

See `../kenya-go-live/02_PRODUCTION_USER_MANUAL.md` for step-by-step workflows.

Related Workspaces

- Shop Floor (</dashboard/shop-floor>) — operator terminal
 - BOM & Formula (</dashboard/bom>) — recipes
 - Recipes (</dashboard/recipes>) — formula definitions
 - Planning (</dashboard/planning>) — MRP and capacity
 - Maintenance (</dashboard/maintenance>) — machine maintenance
 - Quality (</dashboard/quality>) — QC inspections
-

Production Order Status Flow

```

Draft → Released → In Progress → Completed → Closed
      ↓
      Cancelled
  
```

OEE Formula

$OEE = Availability \times Performance \times Quality$

- Availability: planned time vs actual run time
- Performance: actual output rate vs ideal rate
- Quality: good units vs total units produced

Target: $OEE > 85\%$ for most FMCG lines.

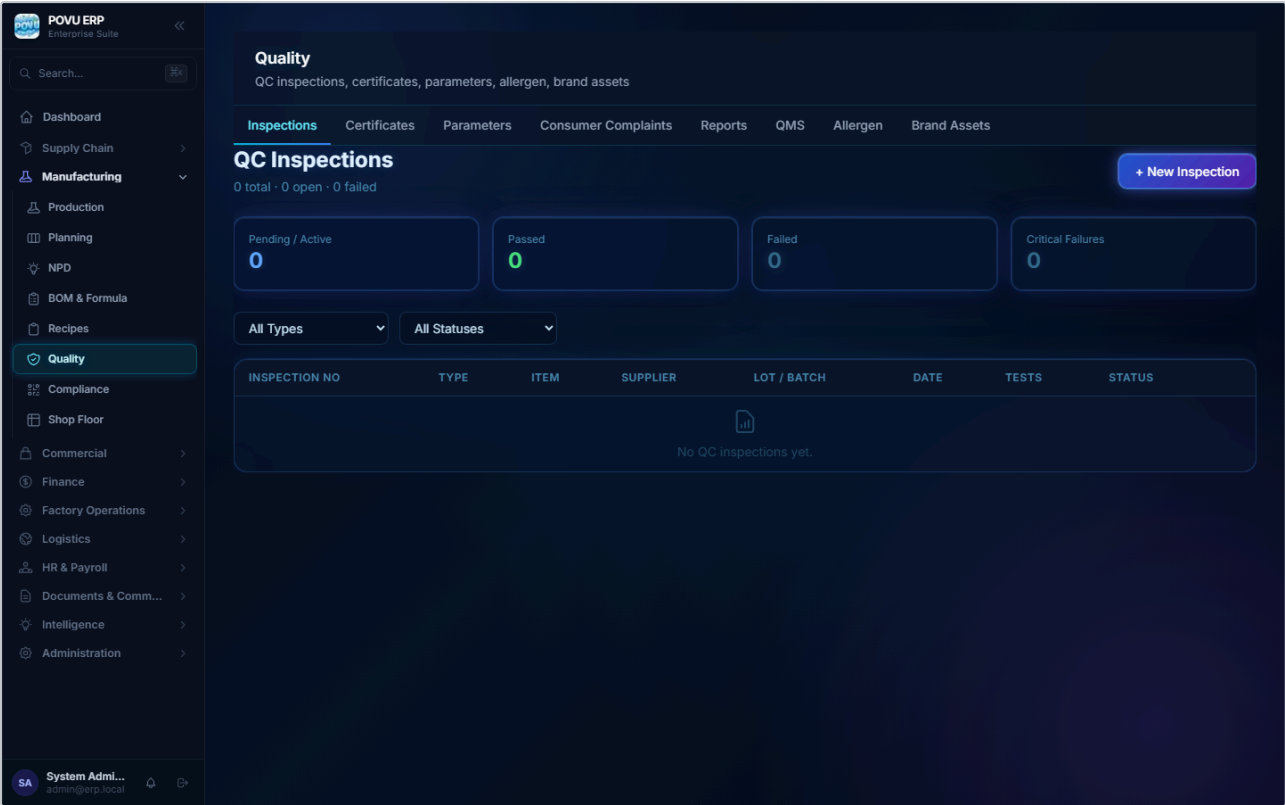
Quality and Compliance

URLs: `/dashboard/quality`, `/dashboard/compliance`

Module: Quality

Permission: `quality.view`

Screenshot



Quality Workspace Tabs

Tab	URL	Purpose
Inspections	?tab=inspections	All QC inspections
Certificates	?tab=certificates	Certificates of Analysis (CoA)
Parameters	?tab=parameters	Test parameters and spec limits
Consumer Complaints	?tab=consumer-complaints	Complaint register
Reports	?tab=reports	QC summary reports
QMS	?tab=qms	HACCP, CAPA, NCR, SOPs
Allergen	?tab=allergen	Allergen matrix
Brand Assets	?tab=brand-assets	Approved artwork and labels

Compliance Workspace Tabs

Tab	URL	Purpose
GS1	?tab=gs1	GS1 barcode generation
Regulatory Certs	?tab=regulatory-certs	KEBS, import, halal certs

Inspection Types

Type	When	Who
Incoming	On receipt of raw materials	QC technician
In-process	During production at CCPs	Operator + QC
Finished goods	After production, before dispatch	QC technician + lab

QMS Modules

HACCP: Hazard Analysis and Critical Control Points

CAPA: Corrective and Preventive Action

NCR: Non-Conformance Report

SOP: Standard Operating Procedure (document library)

Allergen Declarations

Kenya food regulations require accurate allergen labelling. The allergen matrix cross-references all products against the 14 major allergens (EU standards). The system flags cross-contamination risks.

GS1 Barcode Generation

The system generates:

- GS1-128 barcodes: GTIN + batch + expiry
 - QR codes for product traceability
 - Labels printed via connected thermal printer (Zebra, Honeywell)
-

Related Workspaces

- Production (Quality Control tab) — in-process checks
- Inventory (Traceability tab) — batch trace
- Compliance — GS1 and regulatory

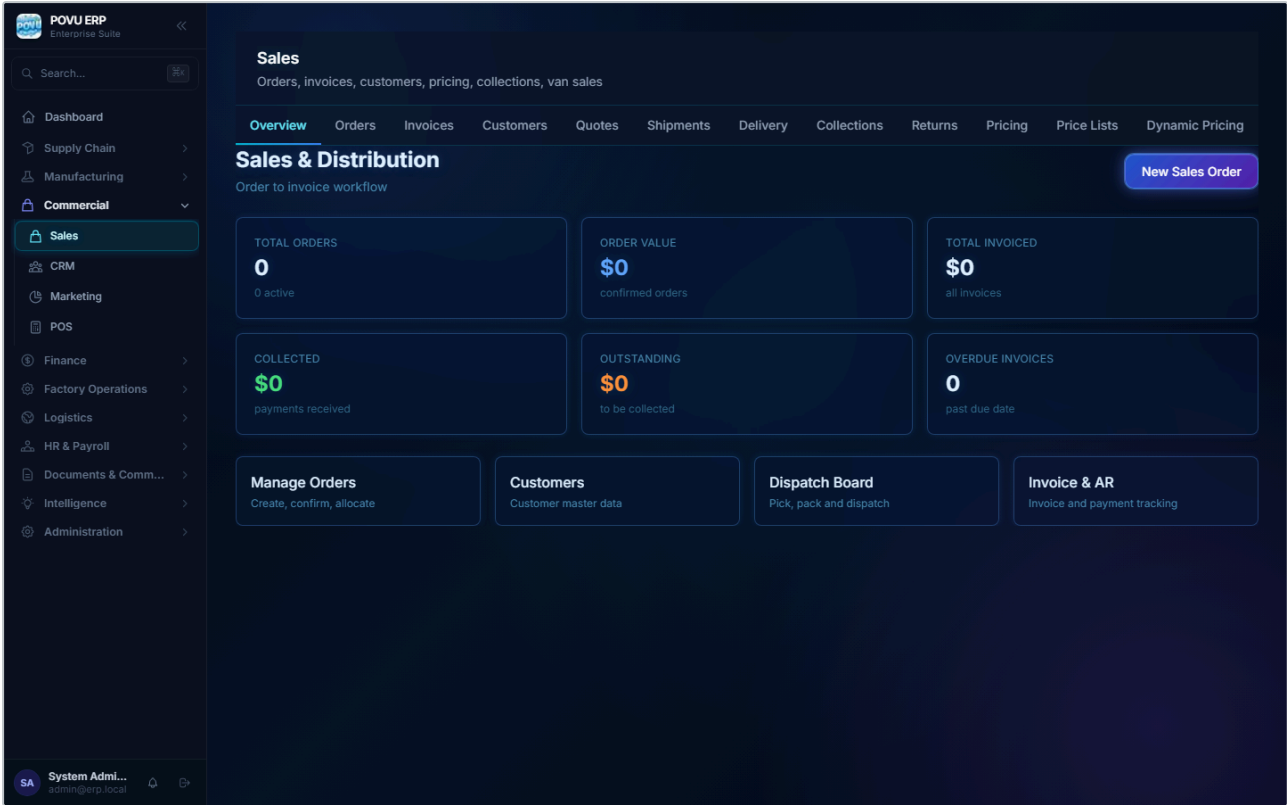
Sales and Distribution

URLs: `/dashboard/sales`, `/dashboard/logistics`, `/dashboard/crm`

Module: Sales

Permission: `sales.view`

Screenshot



Sales Workspace Tabs

Tab	URL	Purpose
Overview	?tab=overview	Sales KPI summary
Orders	?tab=orders	Sales order management
Invoices	?tab=invoices	Customer invoices, eTIMS
Customers	?tab=customers	Customer master
Quotes	?tab=quotes	Quotations
Shipments	?tab=shipments	Delivery notes
Delivery	?tab=delivery	Delivery scheduling
Collections	?tab=collections	Payment collection
Returns	?tab=returns	Return merchandise authorisation
Pricing	?tab=pricing	Base pricing rules
Price Lists	?tab=price-lists	Customer/channel price lists
Dynamic Pricing	?tab=dynamic-pricing	AI-driven pricing
Contracts	?tab=contracts	Customer contracts
Recurring	?tab=recurring	Recurring orders
Commissions	?tab=commissions	Sales rep commissions
Secondary	?tab=secondary	Secondary distribution
Van Sales	?tab=van-sales	Van sales pre-sell / settlement
Distributors	?tab=distributors	Distributor management
Field Sales	?tab=field-sales	Route-to-market tracking
Margin	?tab=margin	Gross margin by order/customer
Portal	?tab=portal	Customer self-service portal
Reports	?tab=reports	Sales reports

Logistics Workspace Tabs

Tab	URL	Purpose
Overview	?tab=overview	Logistics KPI
Shipments	?tab=shipments	Outbound shipments
Containers	?tab=containers	Import container tracking
Arrivals	?tab=arrivals	Expected arrivals
Documents	?tab=documents	Shipping documents
Fleet	?tab=fleet	Vehicle management

CRM Workspace Tabs

Tab	URL	Purpose
Overview	—	CRM KPIs
Pipeline	?tab=pipeline	Sales pipeline board
Leads	?tab=leads	Lead management
Opportunities	?tab=opportunities	Opportunity management
Activities	?tab=activities	Calls, meetings, follow-ups
Forecast	?tab=forecast	Sales forecast
Territory	?tab=territory	Sales territory management
Stages	?tab=stages	Pipeline stage config
Win/Loss	?tab=win-loss	Deal analysis
Loyalty	?tab=loyalty	Loyalty program
NPS	?tab=nps	Net Promoter Score
Surveys	?tab=surveys	Customer surveys

eTIMS (KRA e-Invoicing)

Kenya Revenue Authority requires electronic invoicing for VAT-registered businesses. The system posts each invoice to eTIMS automatically after posting. Requirements:

- Customer must have a KRA PIN on file
 - Product must have a correct VAT code
 - ETIMS credentials configured in Admin → Integrations
-

Van Sales Model

Kenya FMCG distribution uses a van sales model:

1. Pre-selling: sales rep takes orders before loading
 2. Loading: warehouse loads van per approved plan
 3. Selling: driver delivers and collects
 4. Settlement: returns and cash settled at end of route
-

M-Pesa Collections

Collections tab supports M-Pesa as a payment method. M-Pesa transactions are imported via the integration and matched to open invoices.

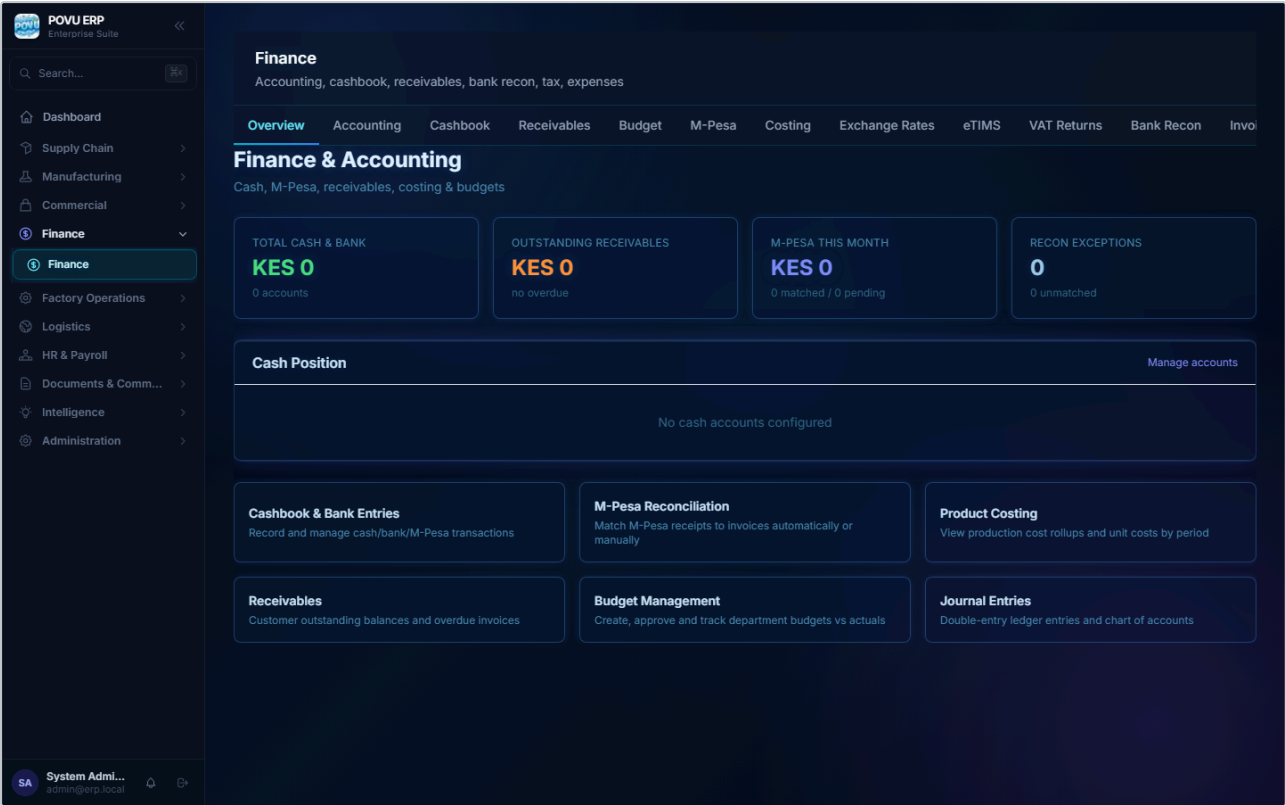
Finance

URL: /dashboard/finance

Module: Finance

Permission: finance.view

Screenshot



Tabs

Tab	URL	Purpose
Overview	—	P&L and cash position summary
Accounting	?tab=accounting	Chart of accounts, journal entries
Cashbook	?tab=cashbook	Cash and petty cash
Receivables	?tab=receivables	Accounts receivable ageing
Budget	?tab=budget	Budget vs actual
M-Pesa	?tab=mpesa	M-Pesa reconciliation
Costing	?tab=costing	Product costing
Exchange Rates	?tab=exchange-rates	FX rate management
eTIMS	?tab=etims	KRA e-invoice status
VAT Returns	?tab=vat-returns	VAT filing preparation
Bank Recon	?tab=bank-recon	Bank statement matching
Invoice Match	?tab=invoice-match	PO/GRN/invoice 3-way match
Fixed Assets	?tab=fixed-assets	Asset register, depreciation
Dimensions	?tab=dimensions	Cost centre / department codes
Dunning	?tab=dunning	Overdue collection letters
Tax	?tab=tax	Tax setup and filing
Bank API	?tab=bank-api	Bank statement auto-import
Expenses	?tab=expenses	Employee expense claims

Kenya-Specific Finance Features

eTIMS: Electronic Tax Invoice Management System — mandatory for VAT-registered companies. See below for current implementation status. **M-Pesa:** Mobile money reconciliation and payment processing
VAT returns: Automated VAT return calculation for KRA filing **Housing Levy:** 1.5% employer + 1.5% employee on gross salary (2024 requirement)

eTIMS Implementation Status

eTIMS fiscalization uses a provider-neutral connector architecture. The connector is complete; live provider calls are not yet active.

Current State

Component	Status
Backend adapter interface (<code>ETIMSCConnector</code> protocol)	Done
<code>SimulationETIMSCConnector</code> (fake ACCEPTED; no network)	Done — active by default
<code>HttpETIMSCConnector</code> skeleton	Skeleton only — auth scheme not yet wired
All fiscalization endpoints (prepare/submit/retry/cancel/poll/health)	Done
Frontend global eTIMS page (<code>/dashboard/finance/etims</code>)	Done — 10-status model, provider health panel, retry/cancel/poll
Frontend eTIMS card in invoice detail	Done — submit/retry/cancel/poll per invoice
<code>production_execution_allowed</code>	<code>false</code> — no live KRA calls
GL posting gate (require ACCEPTED before posting)	Not implemented — blocked on accountant decision
Live provider credentials	Not configured — requires KRA provider selection

eTIMS Status Values

10 statuses: `DRAFT`, `READY`, `PENDING`, `SUBMITTED`, `RETRY_PENDING`, `ACCEPTED`, `REJECTED`, `FAILED`, `ERROR`, `CANCELLED`

What Users Can Do (Simulation Mode)

- View all eTIMS submissions and their status
- Submit, retry, cancel, and poll submissions (all routed to `SimulationETIMSCConnector`)
- Check provider health
- View debug payload details per submission
- Access eTIMS card on individual invoice detail pages

What Is Blocked Until Live Provider Configured

- Real KRA submission and fiscal acceptance

- GL posting gate enforcement
- Production eTIMS control numbers (`control_unit_invoice_no`)

For full workflow and field reference, see the Finance-Payroll module manual: `module-manuals/finance-payroll/07-tax-etims.md`

3-Way Matching

Purchase Invoice → Goods Receipt Note → Purchase Order

All three must match before an invoice is approved for payment. Discrepancies are flagged in Invoice Match tab.

Bank Reconciliation

1. Import bank statement (CSV/OFX) or use Bank API auto-import
2. System matches transactions to ERP entries automatically
3. Review unmatched items manually
4. Post any missing entries
5. Mark reconciliation as complete

Fixed Assets

Full asset lifecycle:

- Capitalization, depreciation (straight-line or reducing balance)
- Asset revaluation
- Disposal and write-off
- Depreciation schedules linked to GL accounts

Related Workspaces

- Sales (Collections tab) — receivables source
- Procurement (Deliveries tab) — payables source
- HR (Payroll) — payroll journal posts to Finance
- Admin (Integrations) — M-Pesa and bank API config

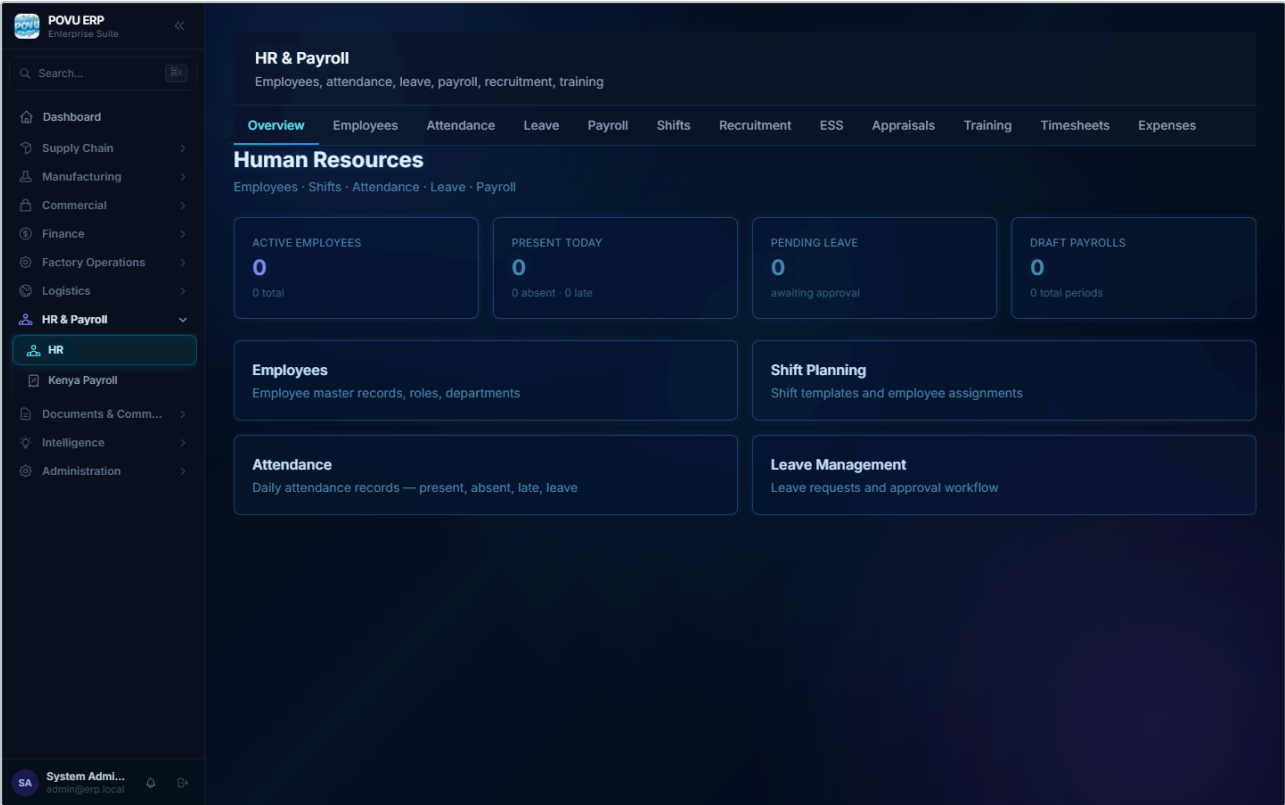
HR and Payroll

URLs: `/dashboard/hr`, `/dashboard/payroll`

Module: HR

Permission: `hr.view`, `payroll_ke.view`

Screenshot



HR Workspace Tabs

Tab	URL	Purpose
Overview	—	HR dashboard
Employees	?tab=employees	Employee master
Attendance	?tab=attendance	Time and attendance
Leave	?tab=leave	Leave management
Payroll	?tab=payroll	Payroll run (shortcut to Payroll workspace)
Shifts	?tab=shifts	Shift scheduling
Recruitment	?tab=recruitment	Hiring pipeline
ESS	?tab=ess	Employee Self-Service portal
Appraisals	?tab=appraisals	Performance reviews
Training	?tab=training	Training records
Timesheets	?tab=timesheets	Time-based billing/tracking
Expenses	?tab=expenses	Expense claims

Kenya Payroll Workspace Tabs

Tab	URL	Purpose
Overview	—	Payroll run summary
Profiles	?tab=profiles	Employee payroll configs
Reports	?tab=reports	P9, NHIF, NSSF, payslips

Kenya Statutory Obligations

Deduction	Rate	Frequency
PAYE	Progressive (KRA bands)	Monthly
NHIF	Per income bracket	Monthly
NSSF	6% employee + 6% employer	Monthly
Housing Levy	1.5% + 1.5%	Monthly

All filed via:

- PAYE: KRA iTax portal
- NHIF: NHIF portal
- NSSF: NSSF portal

ESS (Employee Self-Service)

Employees can:

- View their payslips
- Apply for leave
- Submit expense claims
- Update personal details

Access via `/dashboard/hr?tab=ess` (admin/HR view) or employee-facing portal.

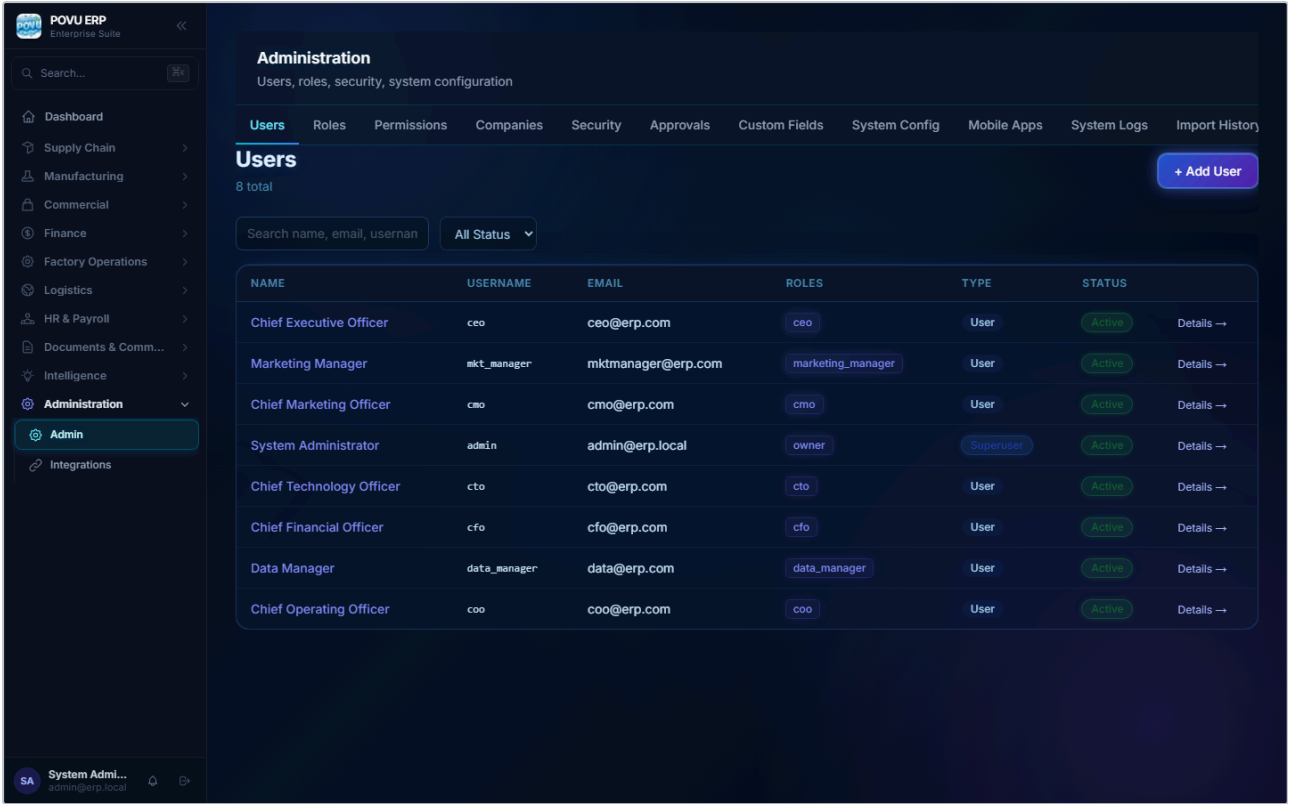
Related Workspaces

- Finance (Expenses tab) — expense claims cross-posted
- Admin (Users tab) — employee users linked to HR records

Admin and Security

URL: `/dashboard/admin` **Module:** Admin **Permission:** `users.view` (admin role)

Screenshot



Admin Workspace Tabs

Tab	URL	Purpose
Users	?tab=users	User account management
Roles	?tab=roles	Role definitions
Permissions	?tab=permissions	Permission matrix
Companies	?tab=companies	Multi-company setup
Security	?tab=security	2FA, password policy, sessions
Approvals	?tab=approvals	Approval workflow rules
Custom Fields	?tab=custom-fields	Add fields to any record type
System Config	?tab=system-config	Company settings, integrations
Mobile	?tab=mobile	Mobile app configuration
Logs	?tab=logs	Audit trail
Import History	?tab=import-history	Bulk data import records

Integrations Workspace Tabs

Tab	URL	Purpose
Overview	—	Integration status
M-Pesa	?tab=mpesa	M-Pesa API credentials
Sync	?tab=sync	Data sync jobs
Marketplace	?tab=marketplace	Third-party app marketplace
Barcode	?tab=barcode	Barcode scanner config
Marketing Sync	?tab=marketing-sync	Marketing tool sync
Logs	?tab=logs	Integration event logs
Webhooks	?tab=webhooks	Outbound webhook endpoints
Developer	?tab=developer	API keys, developer portal

Logs Workspace

URL: `/dashboard/logs`

Tab	Purpose
Operational	General system event log
M-Pesa	M-Pesa API call log
Security	Login attempts, permission changes

Approvals Workspace

URL: `/dashboard/approvals`

Tab	Purpose
All	All pending and completed approvals
Rules	Approval rule configuration

2FA Configuration

Admin → Security tab:

- Enable 2FA requirement per role
- Supported methods: TOTP (authenticator app), Email OTP, SMS OTP
- OTP expiry: 5 minutes (configurable)
- OTP resend: 60 second cooldown

Production guard: `OTP_DEV_DELIVERY_MODE=true` is blocked in production — ensures real OTP delivery via SMTP/Twilio.

Custom Fields

Add organisation-specific fields to any record type without code changes:

- Field types: text, number, date, dropdown, checkbox
- Assign to: products, customers, suppliers, employees, etc.
- Visible in: record forms and list columns

Role-Based Access Control

Permission strings follow the pattern: `{module}.{action}` Examples: `production.view`, `production.create`, `finance.approve`

Each role has a set of permissions. Users can have multiple roles.

Management User Seeding (No Hardcoded Admin Secrets)

Management users (superusers, initial admin accounts) are seeded via environment variables — not hardcoded credentials.

How It Works

The seed script reads management user configuration from env vars at startup:

Variable	Description
<code>MANAGEMENT_USER_EMAIL</code>	Email address for the management superuser
<code>MANAGEMENT_USER_USERNAME</code>	Username
<code>MANAGEMENT_USER_PASSWORD</code>	Initial password (hashed at seed time)
<code>MANAGEMENT_USER_FULL_NAME</code>	Display name
<code>SEED_DEMO_DATA</code>	<code>true</code> to run the full demo seed (products, inventory, production, etc.)

Security rules:

- Never commit actual management credentials to git
- `.env.production` and `.env.development` are gitignored by design
- Only example files (`.env.*.example`) are tracked — these contain no real credentials
- If `MANAGEMENT_USER_PASSWORD` is not set, the seed exits without creating a management user

Force Password Change

Management users created via seed are flagged for forced password change on first login. The user is redirected to the password change page before accessing any other module.

This ensures the seeded initial password is replaced with a user-chosen password before normal operations begin.

PyJWT Authentication

The ERP uses PyJWT for JSON Web Token generation and validation. Key security behaviors:

Behavior	Detail
Token signing	HS256 algorithm with <code>SECRET_KEY</code> from env
Token expiry	Configured via <code>ACCESS_TOKEN_EXPIRE_MINUTES</code>
Expiry check	Token is rejected if <code>exp</code> claim is in the past
Refresh token	Separate <code>REFRESH_TOKEN_EXPIRE_DAYS</code> config
Invalid token	Returns HTTP 401 with <code>WWW-Authenticate: Bearer</code> header
<code>python-jose</code>	Not used — replaced by PyJWT. Do not add <code>python-jose</code> back.

Security note: The `SECRET_KEY` must be a long random string (minimum 32 characters). Set it in `.env.production` — never hardcode it or commit it to git. Generate with: `openssl rand -hex 32`

GS1 Label Generator — Auth-Gated

The GS1 label generator at `/dashboard/compliance/gsl` requires authentication. All 38 GS1 API endpoints are protected by `require_auth` middleware.

Before go-live:

- Unauthenticated access to GS1 endpoints returns HTTP 401
- The frontend GS1 page requires a valid session (login-gated)
- GS1 label printing, template management, and barcode generation all require `users.view` minimum permission

Why this matters: GS1 barcodes are used on product packaging. Unauthorized label generation would create fraudulent barcodes. The auth gate prevents unauthenticated access to the label workflow.

Alembic Migration Safety — PostgreSQL Advisory Lock

Database migrations use a PostgreSQL advisory lock (`pg_advisory_lock(20260517)`) to prevent race conditions in multi-replica deployments.

How It Works

When the backend starts, Alembic runs `upgrade head`. Before running any migration:

1. Attempts to acquire `pg_advisory_lock(20260517)` — a session-level exclusive lock
2. If the lock is held by another replica, waits until released
3. Runs migration once the lock is acquired
4. Releases the lock when migration completes (or on connection close)

Result: Only one replica runs migrations at a time. Other replicas wait. After the first replica completes, subsequent replicas see the DB is already at `head` and skip.

Deployment implication: Rolling deploys are safe — no split-brain migration state. Do not interrupt the startup container before migrations complete; the lock is held until the migration transaction commits.

Blocked Live Integrations

The following integrations are implemented with a simulation/sandbox mode. **Do not enable live mode without the specified prerequisites:**

Integration	Current state	Prerequisites to enable live
eTIMS (KRA fiscalization)	SimulationETIMSCConnector — no KRA calls	Provider selection + KRA sandbox credentials + accountant GL gate approval
M-Pesa (Daraja)	Simulation — no real STK Push	Safaricom Daraja credentials (Consumer Key, Secret, Shortcode, Passkey, Callback URL)
WhatsApp	Demo mode — no Meta Cloud API calls	Meta Cloud API token + Phone Number ID configured in DB
AI (LLM features)	Mock mode — canned responses	Anthropic/OpenAI/Gemini API key in env
SMTP (email/OTP)	Dev delivery mode may be active	Confirm <code>OTP_DEV_DELIVERY_MODE=false</code> in production

Do not enable any live integration without real credentials from the respective provider. Do not commit credentials to git.

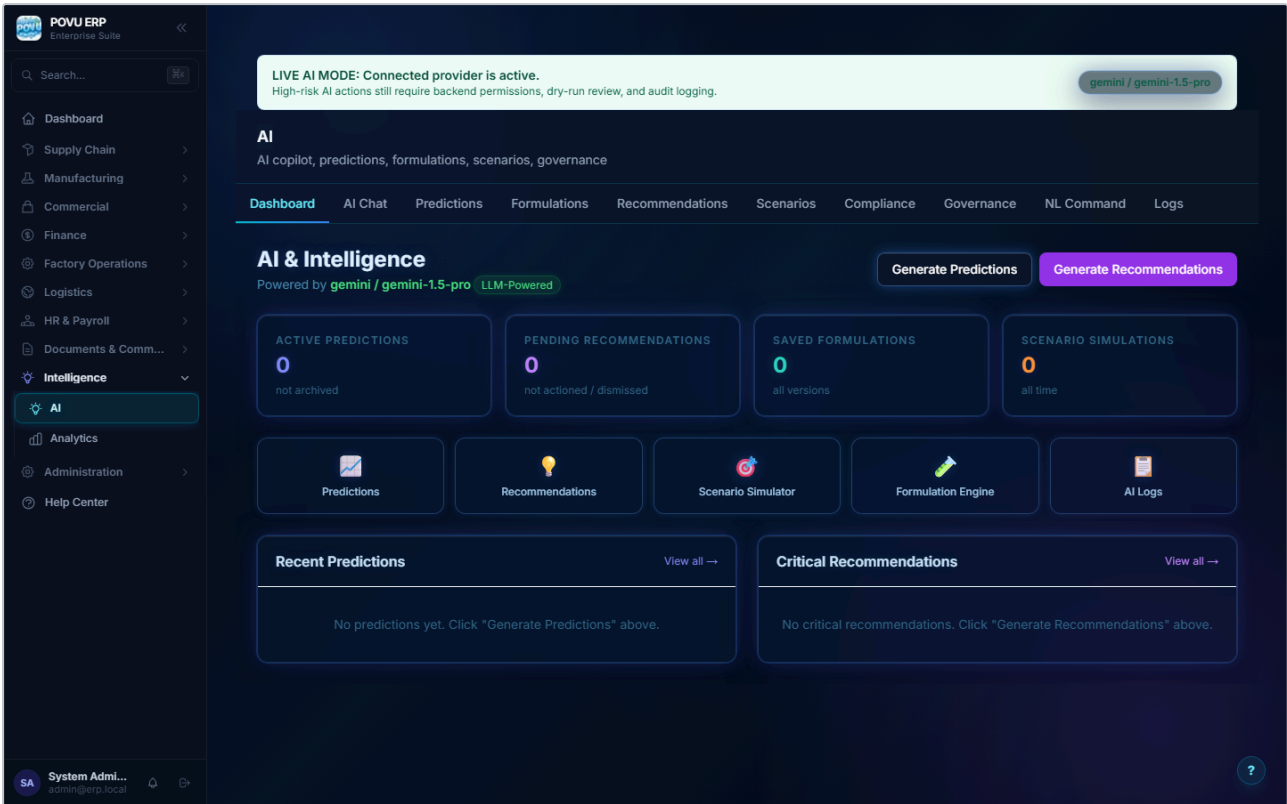
Security Best Practices

- Use strong passwords (12+ chars, mixed case, digits, special chars)
- Enable 2FA for all admin accounts
- Review audit logs weekly
- Deactivate accounts immediately when staff leave
- Never share login credentials
- Rotate `SECRET_KEY` if compromised — all existing tokens become invalid immediately
- Set `OTP_DEV_DELIVERY_MODE=false` in production
- Verify `production_execution_allowed` for all integrations before go-live

AI and Automation

URL: `/dashboard/ai` Module: AI Permission: `ai.view`

Screenshot



AI workspace showing demand forecasting and PROPHET model type (implemented as local Holt-Winters).

AI Workspace Tabs

Tab	URL	Purpose
Dashboard	?tab=dashboard	AI overview and usage stats
Chat	?tab=chat	AI Copilot — natural language queries
Predictions	?tab=predictions	Demand forecasting
Formulations	?tab=formulations	AI-assisted recipe optimisation
Recommendations	?tab=recommendations	Procurement and production suggestions
Scenarios	?tab=scenarios	What-if simulation
Compliance	?tab=compliance	AI-assisted compliance check
Governance	?tab=governance	AI usage policy
NL Command	?tab=nl-command	Natural language ERP commands
Logs	?tab=logs	AI API call log

AI Mode — Mock vs Live

The ERP has two AI tiers:

Tier	Requires	Features
RULE_BASED / STATISTICAL	No external API key needed	Demand forecasting, procurement calculations, production scheduling, payroll
LLM_POWERED	<code>ANTHROPIC_API_KEY</code> or <code>OPENAI_API_KEY</code> required	AI Chat, Formulations, Scenarios, NL Commands

RULE_BASED and STATISTICAL features work without any API key. Demand forecasting uses a local statsmodels implementation — no external service required.

When no LLM API key is set, `AI_PROVIDER=auto` falls back to MockProvider. The `AIModeBanner` shows an amber "MOCK MODE" badge. LLM-powered features return canned demo responses in mock mode.

To enable live LLM features: Set `ANTHROPIC_API_KEY` (recommended), `OPENAI_API_KEY`, or `GEMINI_API_KEY` in `.env.production` and restart the backend. The banner turns green ("LIVE AI MODE") automatically.

AI Copilot (Chat)

Requires LLM API key (`LLM_POWERED` tier).

Ask questions in plain English:

- "What is the current stock of flour?"
- "Show me this week's production orders"
- "Which customers are overdue on invoices?"

AI translates to ERP queries and returns results. Responses are labelled as AI-generated. In mock mode, responses are pre-canned demo text.

Natural Language Commands

The NL Command tab allows admin-enabled users to run ERP operations via natural language. Requires `AI_NL_COMMAND_EXECUTION_ENABLED=true` in environment config.

Disabled by default — enable only after user training. Requires LLM API key.

Demand Forecasting (PROPHET — Local Holt-Winters)

Model type in UI: `PROPHET` **Implementation:** Local statsmodels Holt-Winters exponential smoothing
External API required: No

The `PROPHET` forecast model type in the ERP uses a local Python statsmodels implementation — not Meta's Prophet library and not any external AI API. No API key is required for forecasting.

How It Works

```

Historical sales data
    ↓
Holt-Winters triple exponential smoothing (statsmodels)
    ↓
Forecast lines per SKU per period
    ↓
DemandForecast + DemandForecastLine records stored in DB
    ↓
MRP consumes forecasts for demand-driven planning
  
```

Fallback behaviour: If a product has fewer than the minimum required data points, the model falls back to simple exponential smoothing. If exponential smoothing also fails (e.g. single data point), the system falls back to the historical mean.

Forecast Quality

History available	Behaviour
3+ months of sales data	Full Holt-Winters with trend and seasonality
1–2 months	Exponential smoothing fallback (no seasonality decomposition)
Single data point	Historical mean fallback
No history	No forecast generated

Demand Forecast Seed Data (I6)

The system ships with seeded demand forecasts for all 5 products. These use

`ForecastModelType.PROPHET` with status `COMPLETED` and period type `MONTHLY`. They feed the seeded MRP run (`MRP-SEED-001`).

Tests

Holt-Winters forecasting has a test suite in `backend/tests/`. Tests confirm:

- PROPHET model type maps to local statsmodels (no external API calls)
- Fallback to exponential smoothing when insufficient history
- Fallback to mean when exponential smoothing fails
- Forecast lines generated correctly per product

MRP (Material Requirements Planning)

MRP uses demand forecasts to calculate material requirements and generate procurement suggestions.

MRP run `MRP-SEED-001` is seeded with a 90-day planning horizon covering all 5 products. `POVU-HS` has a `shortage_flag = true` indicating projected stock-out within the planning horizon.

See the Supply Chain manual for full MRP reference.

Known Limitations

Feature	Status
AI Chat	LLM_POWERED — requires ANTHROPIC_API_KEY or OPENAI_API_KEY; mock responses without keys
Demand forecasting	STATISTICAL — no external API needed; local Holt-Winters; needs 3+ months sales history for full model
NL command execution	LLM_POWERED — disabled by default; admin must enable; requires LLM API key
Formulation AI	LLM_POWERED — experimental; review all AI suggestions before applying
Scenarios	Simulation only — does not affect live data
PROPHET enum	Maps to local statsmodels Holt-Winters — not Meta Prophet library; no pip install of prophet required

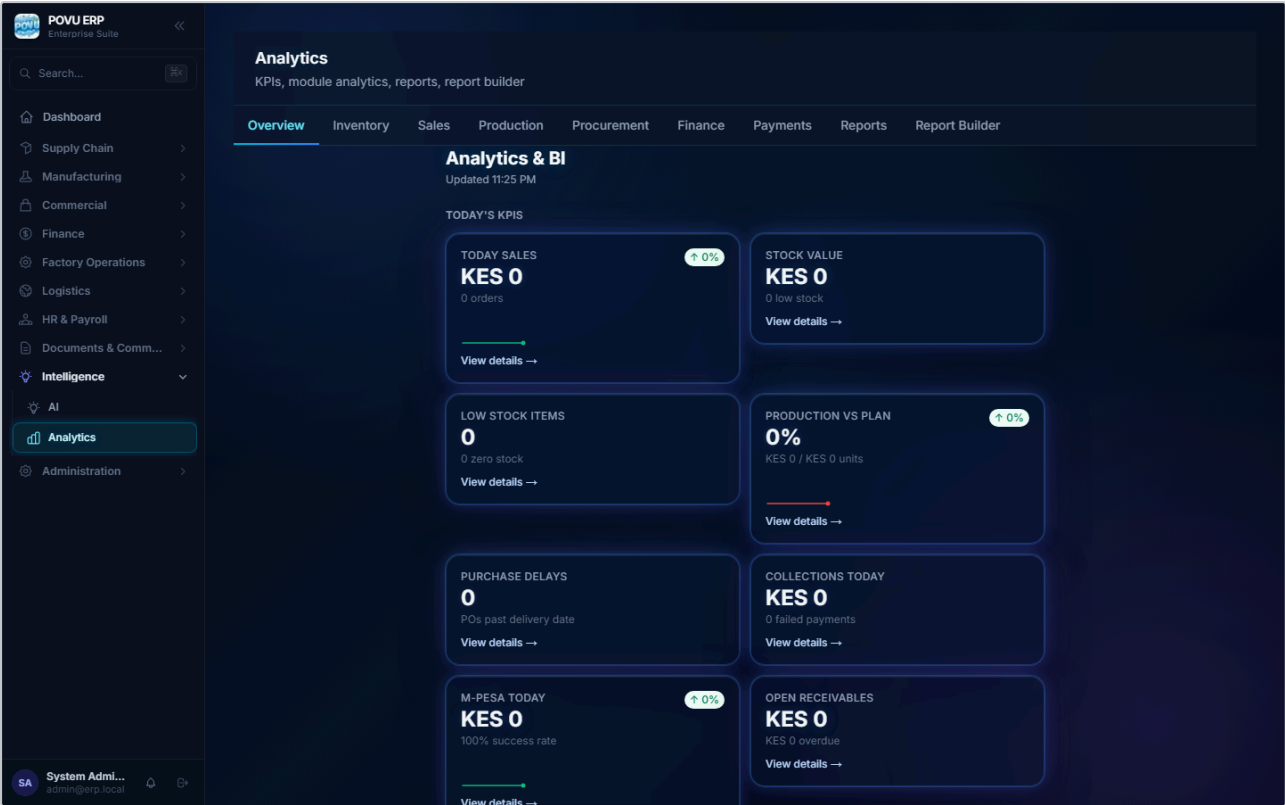
Reports and Exports

URL: /dashboard/analytics

Module: Analytics

Permission: analytics.view

Screenshot



Analytics Workspace Tabs

Tab	URL	Purpose
Overview	?tab=overview	Cross-module KPI dashboard
Inventory	?tab=inventory	Inventory analytics
Sales	?tab=sales	Sales performance
Production	?tab=production	Production and OEE
Procurement	?tab=procurement	Procurement analytics
Finance	?tab=finance	Financial analytics
Payments	?tab=payments	Payment trends
Reports	?tab=reports	Saved reports library
Report Builder	?tab=report-builder	Custom report creator

Built-In Reports

Each workspace has an Export button (CSV/Excel/PDF). Built-in analytics reports:

- Sales by customer/product/region
- Inventory valuation
- Production output vs plan
- Procurement spend by supplier
- Payroll summary
- P&L summary

Report Builder

Create custom reports without coding:

1. Analytics → Report Builder tab
2. Click + **New Report**
3. Select data source
4. Choose columns (dimensions and metrics)
5. Add filters
6. Preview data
7. Save report with a name

8. Schedule: export via email (daily/weekly)

Export Formats

Format	Use Case
CSV	Data processing in Excel
Excel (.xlsx)	Formatted spreadsheet
PDF	Printable report

All exports include the current filter state. Filters visible in the export header.

Scheduled Reports

Currently via manual export. Scheduled email delivery (future feature in Analytics).

Kenya-Specific Reports

Report	Where
P9 (annual PAYE)	HR → Payroll → Reports
NHIF monthly schedule	HR → Payroll → Reports
NSSF monthly schedule	HR → Payroll → Reports
VAT return workings	Finance → VAT Returns
eTIMS submission log	Finance → eTIMS tab

Standalone Operational Pages

Date: 2026-05-18

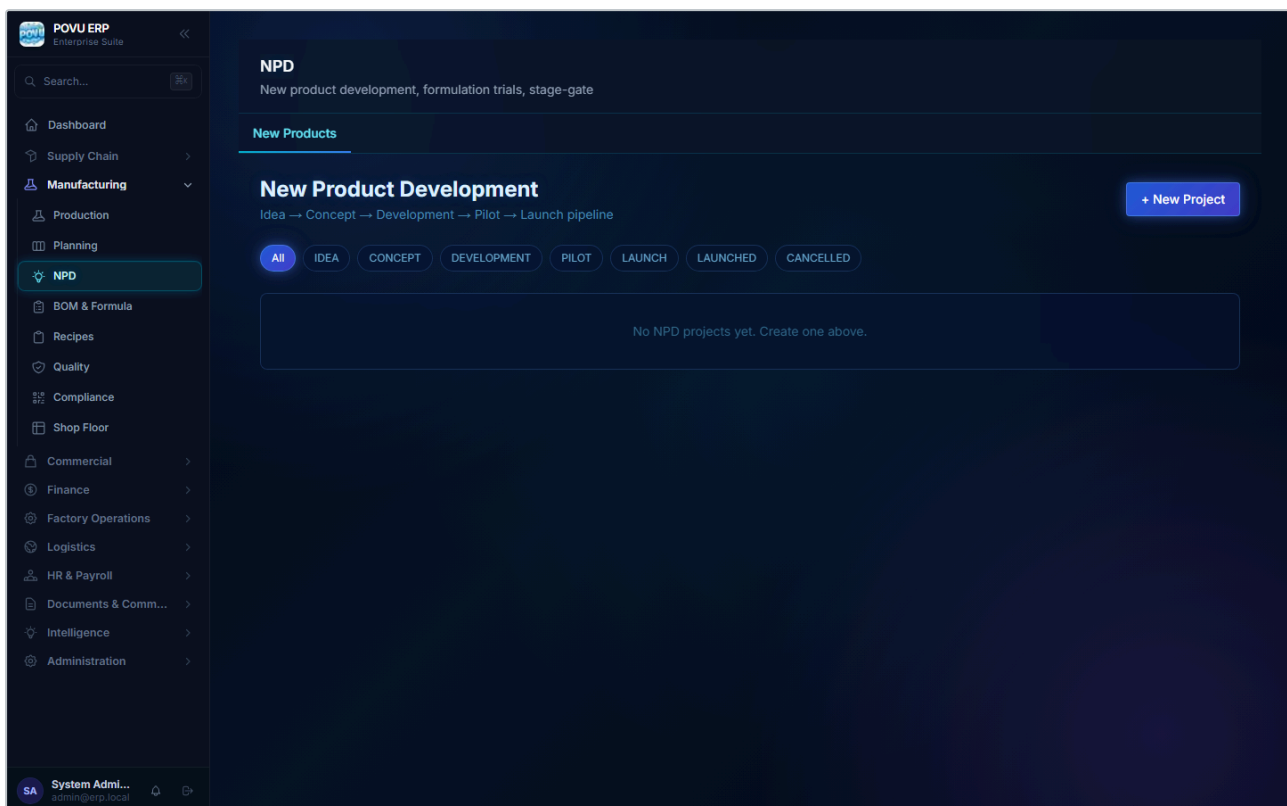
This chapter covers workspaces that serve specific operational functions and do not fit neatly into the main module clusters.

NPD (New Product Development)

URL: `/dashboard/npd`

Tab: npd

Permission: `npd.view`



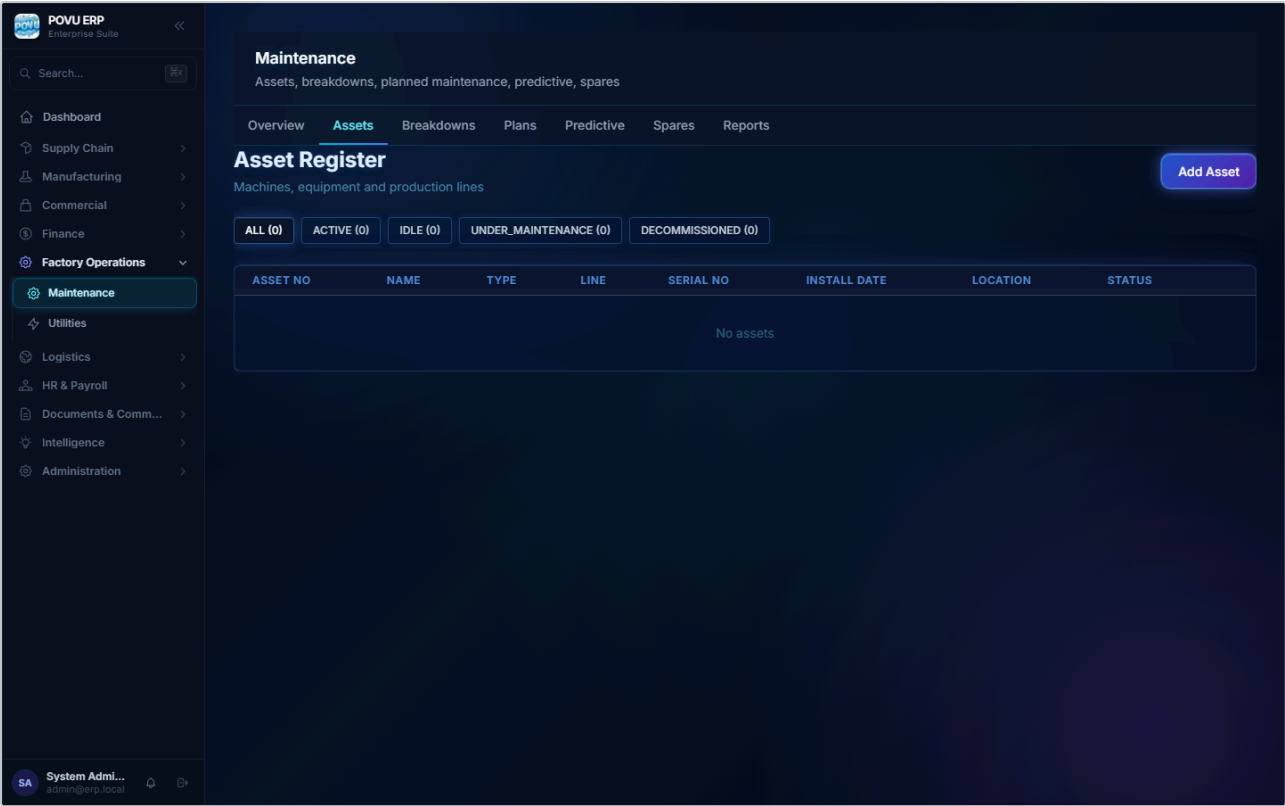
Manages new product development projects from concept to launch:

- Project pipeline with stage gates
- Formula development linked to BOM
- Regulatory approval tracking
- Launch readiness checklist

Maintenance

URL: `/dashboard/maintenance`

Permission: `maintenance.view`



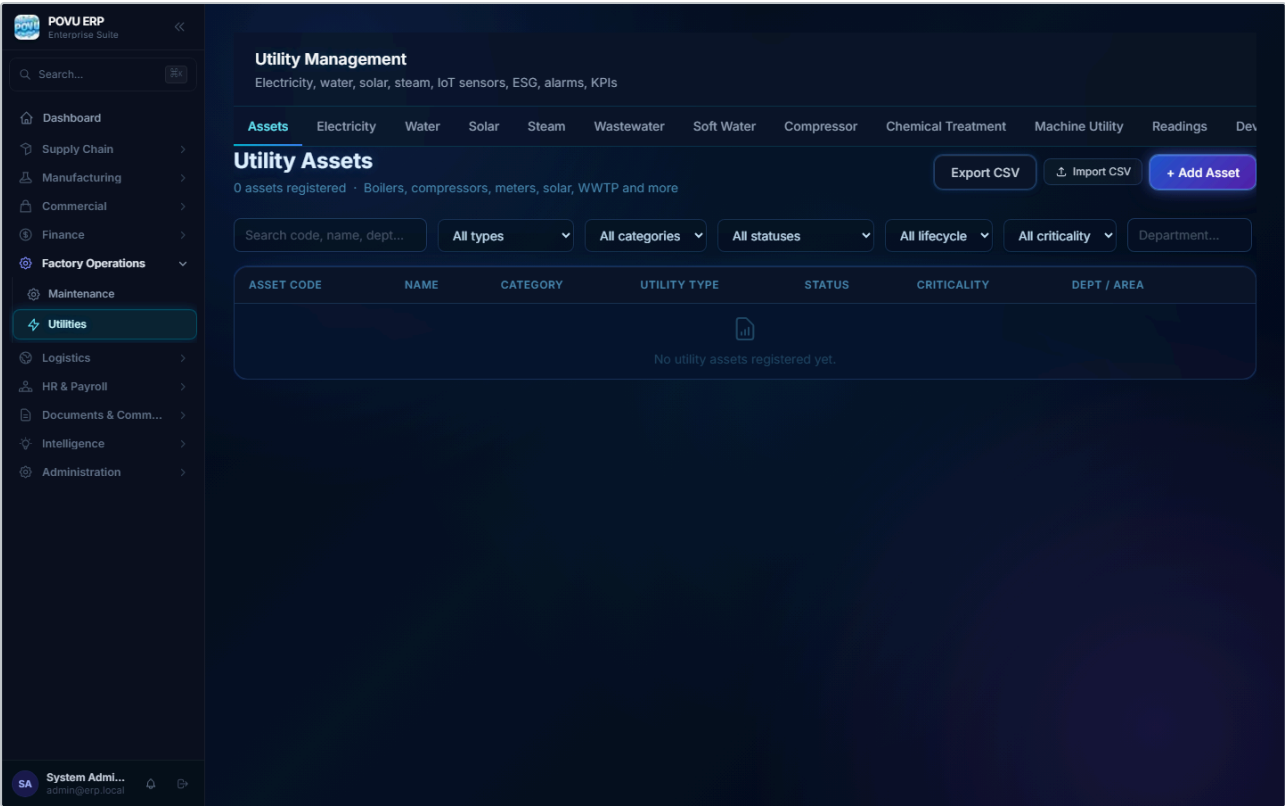
Tab	Purpose
Overview	Maintenance KPI summary
Assets	Equipment register
Breakdowns	Corrective maintenance jobs
Plans	Preventive maintenance schedules
Predictive	Predictive maintenance (sensor data)
Spares	Spare parts inventory
Reports	Maintenance reports

Linked to Production OEE — downtime in Production links to Maintenance breakdown records.

Utility Management

URL: `/dashboard/utility-management`

Permission: `utility_management.view`

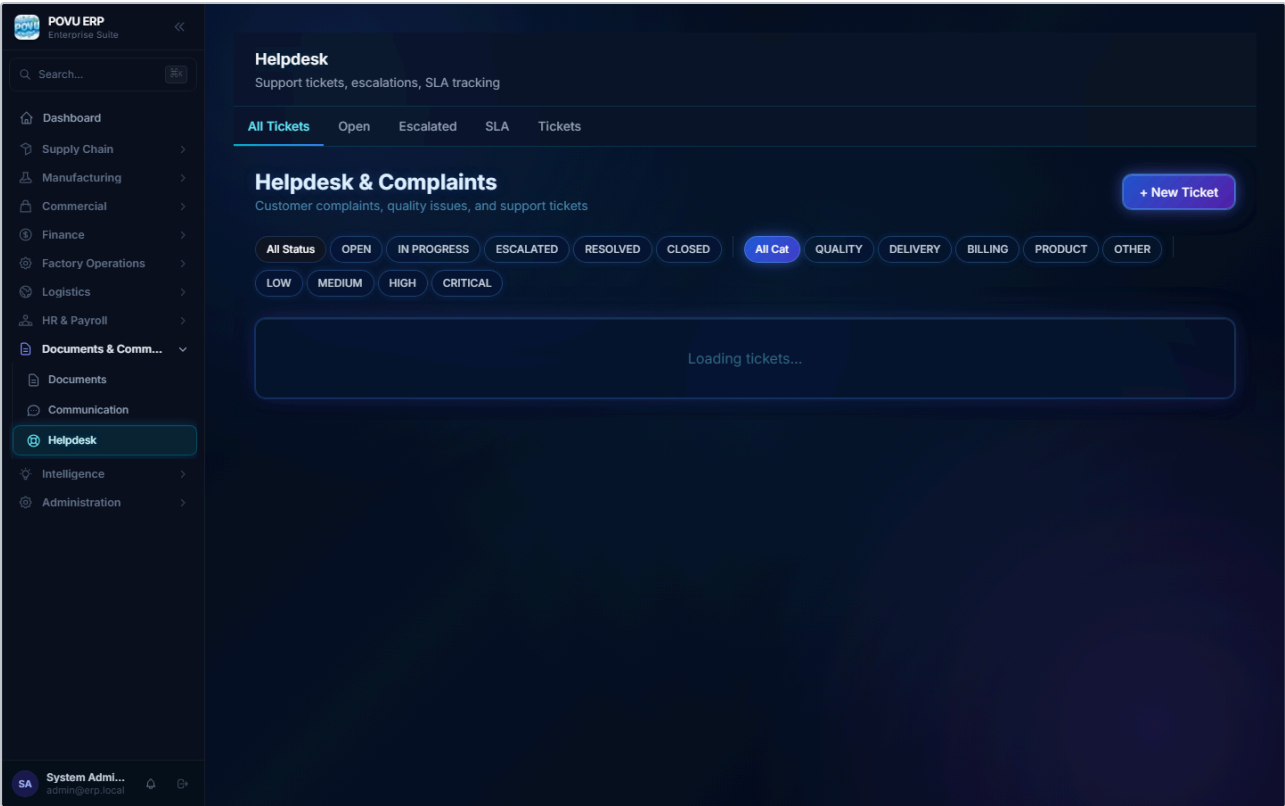


Tab	Purpose
Assets	Utility assets (meters, generators)
Electricity	kWh readings, cost, alerts
Water	Water consumption per process
Solar	Solar generation vs consumption
Steam	Steam boiler readings
Wastewater	Effluent discharge monitoring
Soft Water	Softener readings
Compressor	Air compressor records
Chemical Treatment	Water treatment records
Machine Utility	Per-machine utility allocation
Readings	Manual meter entry
Devices	IoT device configuration
KPI Center	Utility KPI dashboard
Alarm Center	Active alarms
Alarm Rules	Alarm threshold setup
Billing	Utility billing
Transactions	Utility cost transactions
Reports	Utility reports
Integration	External system sync
IoT	IoT stream config
ESG	Environmental metrics for ESG report

Helpdesk

URL: `/dashboard/helpdesk`

Permission: `quality.view`

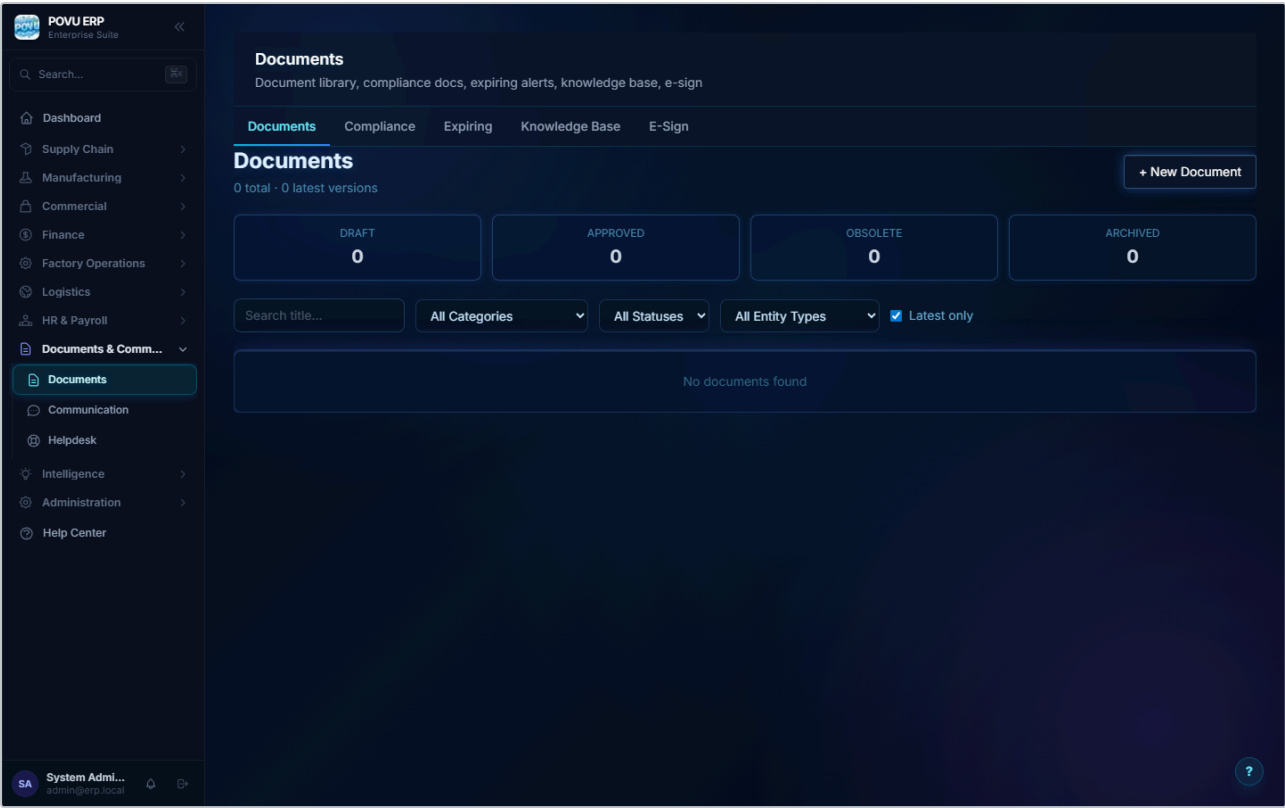


Tab	Purpose
All	All tickets
Open	Open/unresolved tickets
Escalated	Escalated tickets
SLA	SLA compliance
Tickets	Ticket list

Used for internal IT support and customer service tickets. Links to Consumer Complaints (Quality workspace).

Documents

URL: /dashboard/documents Permission: documents.view



Documents workspace showing document library with permission-gated upload and management actions.

Tab	Purpose
Documents	General document library
Compliance	Compliance documents
Expiring	Documents expiring soon
Knowledge Base	SOPs, guides, FAQs
E-Sign	Electronic signature queue

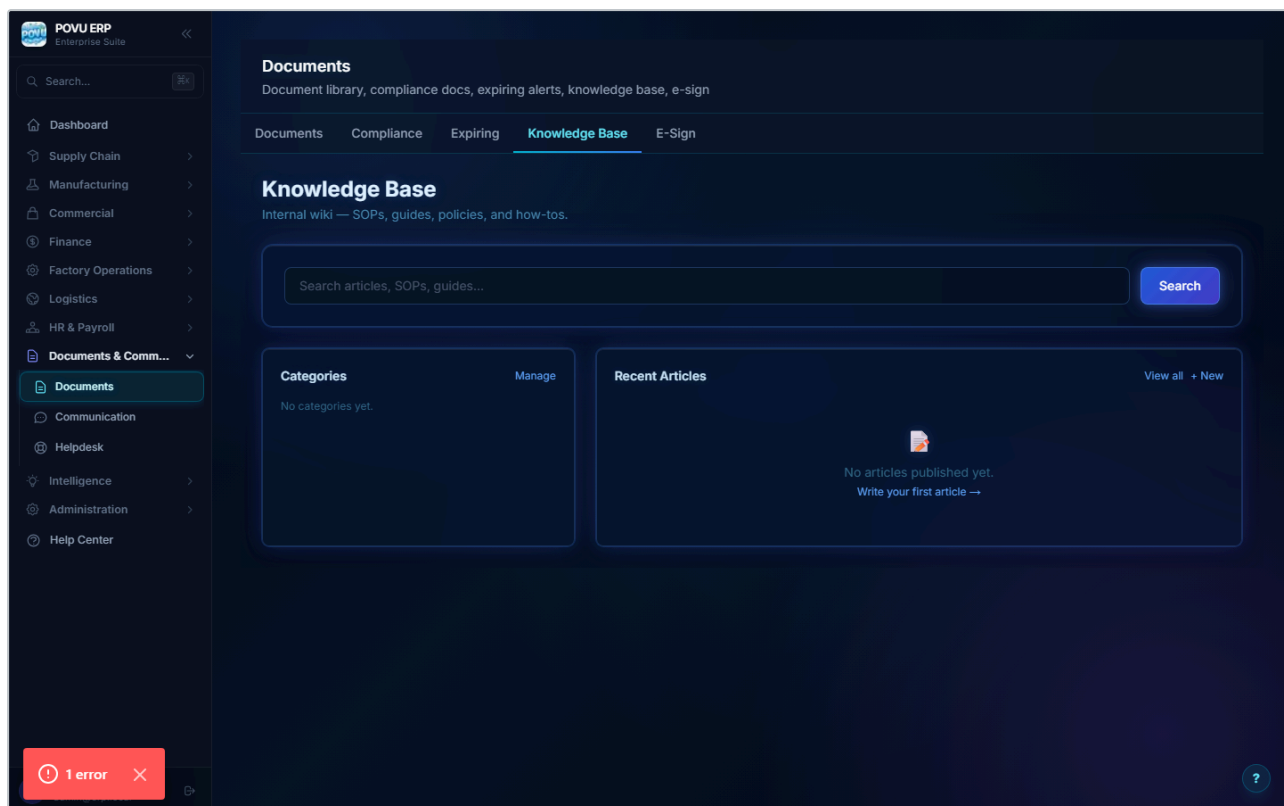
Document Management

The documents module stores file metadata (name, category, status, tags, owner). File content storage requires a storage adapter (S3, local, or Azure) — this is not yet configured in the default installation. Documents currently track metadata only.

Knowledge Base — Permission Requirements

The Knowledge Base tab requires dedicated permissions added in TASK-027:

Action	Permission required
View KB articles	<code>knowledge_base.view</code>
Create KB article	<code>knowledge_base.create</code>
Edit KB article	<code>knowledge_base.edit</code>
Delete KB article	<code>knowledge_base.delete</code>



Knowledge Base tab showing permission-gated article list. Users without `knowledge_base.view` see an access-denied message.

Before TASK-027: KB endpoints used `get_current_user` only — any authenticated user could access all KB routes. **After TASK-027:** All KB endpoints require `require_permission("knowledge_base.*")`. Frontend KB navigation is guarded with `RequirePermission("knowledge_base.view")`.

The article creation page at `/dashboard/knowledge-base/articles/new` is wrapped with `RequirePermission("knowledge_base.create")`. Users without this permission see the standard access-denied screen.

E-Sign — Permission Requirements

The E-Sign page at `/dashboard/esign` requires:

Action	Permission required
View signature queue	<code>esign.view</code>
Sign or decline a document	<code>esign.sign</code>
Request a signature	<code>esign.request</code>

Before TASK-027: E-sign request, list, and dashboard endpoints were unguarded. **After TASK-027:** All 4 e-sign endpoints require `require_permission("esign.*")`. Frontend e-sign page is wrapped with `RequirePermission("esign.view")`.

Note: There is no dedicated `/dashboard/esign` route separate from the Documents workspace. The E-Sign tab is nested inside `/dashboard/documents`. The `esign/page.tsx` exists but is accessed via the Documents module. Captures of the Documents page cover the E-Sign access path.

File Upload Limitation

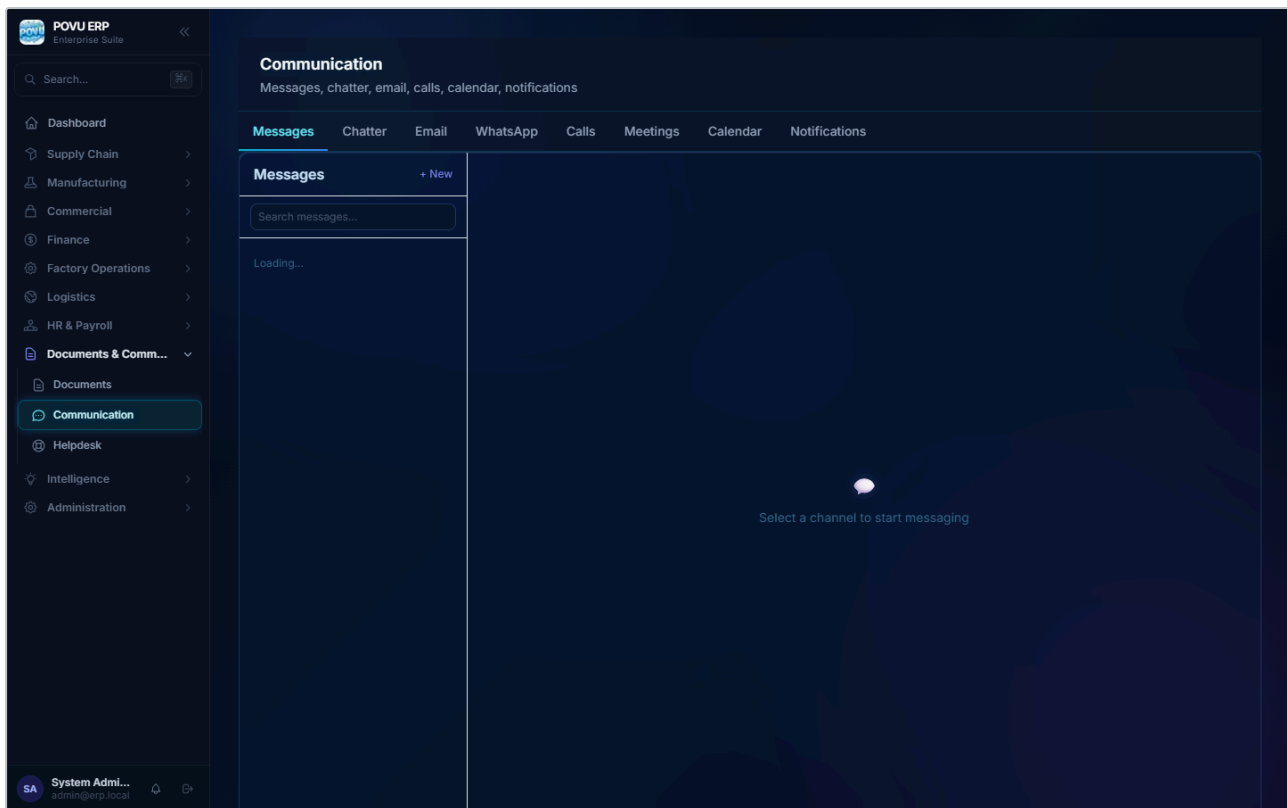
Document file upload (actual file content) is deferred — requires storage adapter configuration (S3 bucket, local volume, or Azure Blob). Until a storage adapter is configured, documents store metadata only. File upload fields in the UI reflect this limitation.

Deferred Items

Feature	Status	Blocker
File storage adapter	Not configured	Storage provider decision (S3/local/Azure)
E-sign cryptographic hash	Not implemented	Security design decision
E-sign expiry automation	Not implemented	Scheduler integration decision
Documents service layer refactor	Low priority	Inline CRUD currently used; no blocker

Communication

URL: `/dashboard/communication`

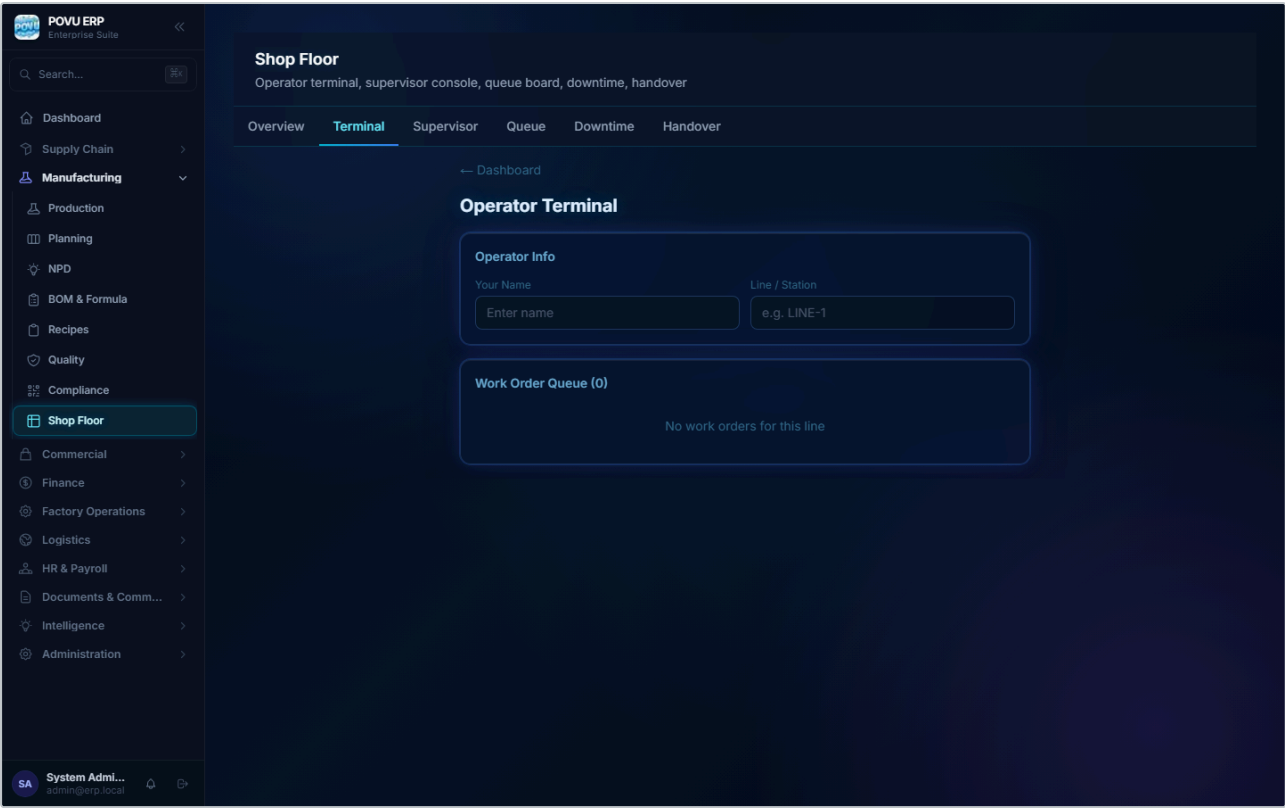


Tab	Purpose
Messages	Internal direct messages
Chatter	Record-linked comments
Email	Email integration
WhatsApp	WhatsApp business messaging
Calls	VoIP call log
Meetings	Meeting scheduler
Calendar	Shared calendar
Notifications	Notification settings

Shop Floor

URL: `/dashboard/shop-floor`

Permission: `production.view`



Tab	Purpose
Overview	Line summary
Terminal	Operator job card confirm/reject
Supervisor	Supervisor floor view
Queue	Work queue board
Downtime	Live downtime logging
Handover	Shift handover notes

Designed for touchscreen use on the factory floor. Large buttons, minimal text.

Old Route Compatibility

Date: 2026-05-18

This chapter documents old URL routes that have been consolidated into workspace tabs. Bookmarks or links using these old URLs will automatically redirect to the correct workspace.

Do NOT capture screenshots for these old routes. They redirect immediately — there is no unique content to document. Instead, document the destination workspace and tab.

How Redirects Work

Old routes redirect via Next.js middleware (308 Permanent Redirect). The browser follows the redirect immediately. The user lands on the correct workspace with the correct tab active.

Source: `frontend/src/lib/routeRedirectMap.ts` and `frontend/src/middleware.ts`.

Old Route → New Workspace Mapping

Sales & Distribution

Old Route	Redirects To	Tab
/dashboard/van-sales	/dashboard/sales	tab=van-sales
/dashboard/sales/orders/[id]	/dashboard/sales	—
/dashboard/portal	/dashboard/sales	tab=portal
/dashboard/customers	/dashboard/sales	tab=customers
/dashboard/contracts	/dashboard/sales	tab=contracts
/dashboard/price-list	/dashboard/sales	tab=price-lists
/dashboard/dynamic-pricing	/dashboard/sales	tab=dynamic-pricing
/dashboard/commissions	/dashboard/sales	tab=commissions
/dashboard/field-sales	/dashboard/sales	tab=field-sales
/dashboard/secondary-sales	/dashboard/sales	tab=secondary
/dashboard/distributors	/dashboard/sales	tab=distributors
/dashboard/returns	/dashboard/sales	tab=returns
/dashboard/delivery	/dashboard/logistics	tab=shipments

Finance

Old Route	Redirects To	Tab
/dashboard/fixed-assets	/dashboard/finance	tab=fixed-assets
/dashboard/bank-reconciliation	/dashboard/finance	tab=bank-recon
/dashboard/invoice-match	/dashboard/finance	tab=invoice-match
/dashboard/tax	/dashboard/finance	tab=tax
/dashboard/expenses (finance)	/dashboard/finance	tab=expenses
/dashboard/dunning	/dashboard/finance	tab=dunning
/dashboard/dimensions	/dashboard/finance	tab=dimensions
/dashboard/finance/accounting	/dashboard/finance	—

Quality

Old Route	Redirects To	Tab
/dashboard/qms	/dashboard/quality	tab=qms
/dashboard/allergen	/dashboard/quality	tab=allergen
/dashboard/shelf-life	/dashboard/inventory	tab=shelf-life
/dashboard/traceability	/dashboard/inventory	tab=traceability
/dashboard/consumer-complaints	/dashboard/quality	tab=consumer-complaints
/dashboard/gs1	/dashboard/compliance	tab=gs1
/dashboard/regulatory-certs	/dashboard/compliance	tab=regulatory-certs

HR

Old Route	Redirects To	Tab
/dashboard/recruitment	/dashboard/hr	tab=recruitment
/dashboard/appraisals	/dashboard/hr	tab=appraisals
/dashboard/timesheets	/dashboard/hr	tab=timesheets
/dashboard/training	/dashboard/hr	tab=training
/dashboard/ess	/dashboard/hr	tab=ess
/dashboard/payroll/runs/[id]	/dashboard/hr	tab=payroll

Production

Old Route	Redirects To	Tab
/dashboard/production/orders	/dashboard/production	—
/dashboard/production/orders/[id]	/dashboard/production	—
/dashboard/production/costing	/dashboard/production	tab=costing
/dashboard/copacking	/dashboard/production	—
/dashboard/mrp	/dashboard/planning	tab=mrp
/dashboard/mps	/dashboard/planning	tab=mps
/dashboard/planning/schedule	/dashboard/planning	tab=schedule

Inventory

Old Route	Redirects To	Tab
/dashboard/cycle-count	/dashboard/inventory	tab=cycle-count
/dashboard/serial-tracking	/dashboard/inventory	tab=serials
/dashboard/inventory/stock	/dashboard/inventory	tab=stock
/dashboard/inventory/movements	/dashboard/inventory	tab=movements
/dashboard/wms (picking)	/dashboard/warehouses	tab=wms

Procurement

Old Route	Redirects To	Tab
/dashboard/procurement/orders/[id]	/dashboard/procurement	—
/dashboard/procurement/purchase-orders	/dashboard/procurement	tab=orders
/dashboard/procurement/rfq	/dashboard/procurement	tab=rfq
/dashboard/subcontracting	/dashboard/procurement	tab=subcontracting
/dashboard/landed-cost	/dashboard/procurement	tab=landed-cost
/dashboard/supplier-portal	/dashboard/procurement	tab=supplier-portal

Admin

Old Route	Redirects To	Tab
/dashboard/users	/dashboard/admin	tab=users
/dashboard/users/[id]	/dashboard/admin	tab=users
/dashboard/roles	/dashboard/admin	tab=roles
/dashboard/audit-logs	/dashboard/admin	tab=logs
/dashboard/company	/dashboard/admin	tab=companies
/dashboard/custom-fields	/dashboard/admin	tab=custom-fields
/dashboard/approvals (admin)	/dashboard/admin	tab=approvals

Analytics & Reports

Old Route	Redirects To	Tab
/dashboard/reports	/dashboard/analytics	tab=reports
/dashboard/report-builder	/dashboard/analytics	tab=report-builder
/dashboard/market-intelligence	/dashboard/marketing	tab=market-intel

CRM & Marketing

Old Route	Redirects To	Tab
/dashboard/crm/records/[id]	/dashboard/crm	—
/dashboard/loyalty	/dashboard/crm	tab=loyalty
/dashboard/nps	/dashboard/crm	tab=nps
/dashboard/surveys	/dashboard/crm	tab=surveys
/dashboard/promotions	/dashboard/marketing	tab=promotions-schemes
/dashboard/tpm	/dashboard/marketing	tab=tpm
/dashboard/brand-assets	/dashboard/quality	tab=brand-assets

Finance (Additional)

Old Route	Redirects To	Tab
/dashboard/payroll-ke	/dashboard/payroll	—
/dashboard/knowledge-base	/dashboard/documents	tab=knowledge-base
/dashboard/esign	/dashboard/documents	tab=esign
/dashboard/meetings	/dashboard/communication	tab=meetings
/dashboard/calendar	/dashboard/communication	tab=calendar
/dashboard/whatsapp	/dashboard/communication	tab=whatsapp

Notes for Manual Authors

When writing manual text about an old route:

"If you previously bookmarked `/dashboard/van-sales`, it now automatically redirects to the Sales workspace Van Sales tab (`/dashboard/sales?tab=van-sales`). Update your bookmark."

Do NOT create a manual chapter for an old redirect-only route. Link to the destination workspace chapter instead.